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IEEE Guide for Identification, Testing, and Evaluation of the Dynamic Performance of Excitation Control Systems

IEEE Power and Energy Society

Sponsored by the
Energy Development and Power Generation Committee

IEEE
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USA

IEEE Std 421.2™-2014
(Revision of
IEEE Std 421.2-1990)

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Approved 27 March 2014

IEEE-SA Standards Board

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Abstract: This guide includes criteria, definitions, and test procedures for evaluating the dynamic performance of excitation control systems for synchronous machines as applied by electric utilities.

Keywords: Closed-loop performance, dynamic performance, excitation control systems, excitation system specifications, IEEE Std 421.2™, large signal disturbances, model, small signal disturbances, stabilizer, synchronous machine

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Introduction

This introduction is not part of IEEE Std 421.2-2014, IEEE Recommended Practice for Overvoltage and Insulation Coordination of Transmission Systems at 1000 kV AC and Above.

This guide presents dynamic performance criteria, definitions, and test objectives for excitation control systems as applied by electric utilities. It should be specifically noted that the term *excitation control system* refers to the entire control system including the synchronous machine and power system as well as the excitation system.

The Working Group on Excitation Control System Dynamic Performance of the Excitation Systems and Controls Subcommittee of the Energy Development and Power Generation Committee adopted many definitions and performance criteria, which are common to all control systems, and derived others specifically related to excitation control systems. In doing this, the material in IEEE Std 421.1, IEEE Standard Definitions for Excitation Systems for Synchronous Machines and the now discontinued IEEE Std 100-1988, IEEE Standard Dictionary of Electrical and Electronics Terms (ANSI) was heavily used. Efforts were made not to conflict with existing definitions and criteria, but to clarify, supplement, and more fully define them as related specifically to excitation control systems. Definitions from these standards, which are reproduced in this standard, are included in Clause 3.

In preparing this guide, the working group recognized that both factory testing and field testing of excitation control systems and some of their components are costly and often impractical. Alternator-rectifier excitation systems in which the terminals of the exciter may not be available may preclude field testing of the exciter separately. Compound source excitation systems whose power is derived from the generator currents and voltages present special difficulties. Providing a load which reasonably duplicates the generator field characteristics, so that regulation effects and proper waveforms can be adequately simulated, may not be economically justifiable. Field tests on units under normal operating conditions are constrained to comply with the operating and security requirements of the power system, which often prohibit large excursions of the excitation control system variables. For many applications, it is necessary to devise practical test procedures for individual components and then by analytical means, to verify the total excitation system performance.

The need for models that accurately simulate the operation of excitation control systems during system disturbances demands effective test methods. However, the practical limitations on tests make it difficult to measure the parameters required for models. One solution is the collection of more complete data during system disturbances. Present instrumentation practices generally do not include the collection of data from enough excitation control system variables to permit the data to be used for model refinement. Improving the quantity and quality of data collected during system disturbances may be the only practical way to obtain the data required to significantly improve the accuracy of large signal models.

This revision incorporates a general introduction to the dynamic performance classification section to aid understanding, and the effects of excitation limiters are introduced. Clarifications have been introduced where technology advancements have changed approaches in modelling, such as the introduction of integral control terms whereby open-loop steady-state gain is considered infinite. Additional and improved figures have been introduced and some typical parameters are shown. It is important to note that this document is part of the IEEE 421 series of guides on the various aspects of excitation control systems for synchronous generators and that the IEEE Excitation Systems and Controls Subcommittee has also produced a tutorial on Power System Stabilization via Excitation Control which provides substantial supplemental material.

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1. Scope

This guide includes criteria, definitions, and test objectives for evaluating the dynamic performance of excitation control systems as applied by electric utilities. The term *excitation control system* (see Figure 1) is used to distinguish the combined performance of the synchronous machine, power system, and excitation system from that of the excitation system alone (see IEEE Std 421.1™). The primary purpose of this guide is to provide a basis for evaluating the closed-loop performance of excitation control systems, including synchronous machines, for both large and small signal disturbances. Confirming the adequacy of mathematical models for excitation control systems for use in analytical studies of power systems, identifying objectives for tests of excitation control systems and their components, and preparing excitation system specifications and additional standards will also be addressed (see IEEE Std 421.3™, IEEE Std 421.4™, and IEEE Std 421.5™,). Portions of this guide will also serve as educational material for people who are becoming familiar with excitation control systems. This guide is applicable to excitation systems used on all sizes and types of synchronous machines including those in nuclear power facilities.

Traditionally, large signal performance (see 4.2) has been more closely associated with equipment specification and acceptance testing, while small signal performance (see 4.3) has been more closely associated with stability and model studies. Matching actual disturbance data with model simulations, however, requires that both large and small signal performance criteria be considered during design, specification, and acceptance testing.

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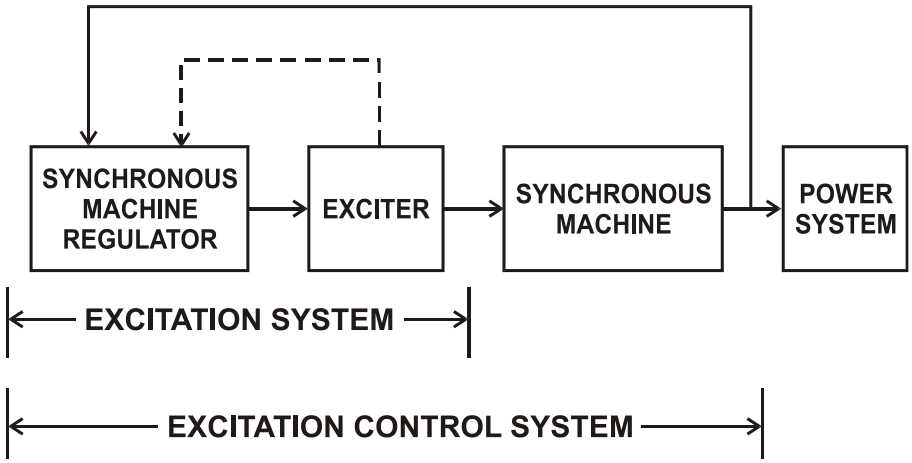

Figure 1—Block diagram of an excitation control system

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 421.1™, IEEE Standard Definitions for Excitation Systems for Synchronous Machines.[1]

IEEE Std 421.3™, IEEE Standard for High-Potential Test Requirements for Excitation Systems for Synchronous Machines.

IEEE Std 421.4™, IEEE Guide for the Preparation of Excitation System Specifications.

IEEE Std 421.5™, IEEE Recommended Practice for Excitation System Models for Power System Stability Studies.

IEEE Std C50.13™, IEEE Standard for Cylindrical-Rotor 50 Hz and 60 Hz Synchronous Generators Rated 10 MVA and Above.

3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.[2] A complete set of excitation system definitions is presented in IEEE Std 421.1™; however, definitions critical for using this guide are repeated here with additional clarification as needed for the specific application. Additionally, some feedback control system terms from are included.

air-gap field current: The synchronous machine field current required to produce rated voltage on the air-gap line of the synchronous machine.

[1] IEEE publications are available from The Institute of Electrical and Electronics Engineers (http://standards.ieee.org/).
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air-gap field voltage: The synchronous machine field voltage required to produce rated voltage on the air-gap line of the synchronous machine with its field winding at 1) 75 °C for field windings designed to operate at rating with a temperature rise of 60 °C or less; or 2) 100 °C for field windings designed to operate at rating with a temperature rise greater than 60 °C.

NOTE—This defines one per unit excitation system voltage for use in computer representation of excitation systems (IEEE Std 421.5™), however, this value may need to be adjusted in certain simulations to match dynamic responses performed with the field winding temperature at other temperatures.[3]

automatic voltage regulator (AVR): This term refers to the controller that regulates the synchronous machine terminal voltage. More recent systems frequently include limiters, field current regulator and power system stabilizer with the AVR within the same electronic module. The term *excitation controller* may be used to refer to such an electronic module to emphasize that the included control algorithms adjust machine excitation with a wider purpose than simple regulation of terminal voltage.

ceiling current: The maximum field current that the excitation system is designed to supply. Typically, this is related to the thermal capability of the excitation system equipment or the synchronous machine field circuit capability.

NOTE—There can be separate short-time and steady-state values.

ceiling voltage: The maximum direct voltage that the excitation system is designed to supply from its terminals under defined conditions.

dynamic response: The output voltage of a control system or intermediate point in a system expressed as a function of time.

excitation control system: The feedback control system that includes the synchronous machine and its excitation system. The term is used to distinguish the performance of the synchronous machine and excitation system in conjunction with the power system from that of the excitation system alone.

excitation limiter: A control function that provides an additional input to the voltage regulator to prevent a current, voltage, or other electrical quantity from exceeding preset limits. Typical excitation limiters include overvoltage (OV), volts/hertz (V/Hz), instantaneous and timed overexcitation limiter (OEL), underexcitation limiter (UEL), and stator overcurrent limiters (SCL). Individual limiters are described in IEEE Std 421.1™.

excitation system nominal response: A figure of merit for non-high initial response excitation systems related to the ability of the excitation system to respond to transient stability issues and to permit comparisons of excitation systems. The nominal response describes the rate of increase of the excitation system output voltage and is determined from the excitation system voltage response curve (see 5.6).

excitation system rated current: The direct current at the excitation system output terminals that the excitation system can supply under defined conditions of its operation. This current is at least that value required by the synchronous machine under the most demanding continuous operating conditions (generally resulting from synchronous machine voltage, frequency, and power factor variations).

excitation system rated voltage: The direct voltage at the excitation system output terminals that the excitation system provides when delivering excitation system rated current under rated continuous load conditions of the synchronous machine with its field winding at (1) 75 °C for field windings designed to operate at rating with a temperature rise of 60 °C or less; or (2) 100 °C for field windings designed to operate at rating with a temperature rise greater than 60 °C.

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excitation system voltage response time: The time in seconds for the excitation voltage to attain 95% of the difference between ceiling voltage and rated field voltage under specified conditions.

frequency response characteristic (linear system): In a linear system, the frequency-dependent relation, in both gain and phase difference, between steady-state sinusoidal inputs and the resultant steady-state sinusoidal outputs.

high initial response (HIR): An excitation system capable of attaining 95% of the difference between ceiling voltage and rated field voltage in 0.1 s or less under specified conditions.

large signal performance: Response of an excitation control system, excitation system, or elements of an excitation system to signals that are large enough that nonlinearities must be included in the analysis of the response to obtain realistic results.

power system stabilizer (PSS): A function that provides an additional input to the voltage regulator to improve the damping of power system oscillations.

NOTE—A number of different quantities may be used as input to the power system stabilizer, such as shaft speed, frequency, electric power, etc., or a combination of these signals.

small signal performance: The response of an excitation control system, excitation system, or elements of an excitation system to signals that are small enough that nonlinearities can be disregarded in the analysis of the response, and operation can be considered to be linear.

4. Dynamic performance classification

4.1 Overall

The subclauses below describe methods for tuning and testing excitation system functions as applied to synchronous generators. The techniques and discussions apply directly to analog electronic proportional style controls with feedback from various measured signals and measured with external testing and recording instruments. Actual excitation system equipment and test equipment, however, have evolved from magnetic and rotating devices, through analog electronic implementations to digital control implementations. Both the controls and the test equipment details must be understood and the techniques described below adapted appropriately to achieve the desired tuning goals. Additionally, modern controls may have multiple modes of operation where gains or feedback signals are changed to allow better performance depending on unit conditions such as start-up, on-line at full load, or shut-down. As a result, tests performed in one operating condition, e.g., off-line, may not capture the control gains and time constants which are in service under load. A thorough review of the control gains and time constants settings as well as logic, which may alter these parameters, must be performed prior to embarking on a testing and tuning program.

4.2 Large signal performance

Large signal performance is the response of an excitation control system, excitation system, or elements of an excitation system to signals that are large enough that nonlinearities must be included in the analysis of the response to obtain accurate results.

The purpose of large signal performance criteria is to provide a means of evaluating excitation system performance for severe transients that may include large variations in synchronous machine stator voltages,

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synchronous machine stator currents, and induced synchronous machine field currents; that is, for transients affecting power system transient stability.

To assess the ability of the excitation system to improve synchronous machine performance, the criteria must reflect the effects of operation under realistic power system disturbances. With respect to performance testing, it is often impractical to adequately duplicate all of these effects.

In cases where tests can only be made on individual components and only at partial load or open circuit, analytical means may be used to predict performance under actual operating conditions. Criteria used to assess the large signal performance include quantities such as dynamic responses, ceiling currents and voltages, voltage response times, and nominal responses derived from dynamic responses (see IEEE Std 421.1™).

4.3 Small signal performance

Small signal performance is the response of an excitation control system, excitation system, or elements of an excitation system to signals that are small enough that nonlinearities can be disregarded in the analysis of the response, and operation can be considered to be linear. Small signal performance of an excitation control system or its components can be assessed from dynamic responses, frequency responses, or by eigenvalue analysis. (See IEEE Std 421.5, Committee Report, 1973 [B6], IEEE Tutorial Course [B23], Kundur [B29], Kundur, et al. [B30], for examples.)

Small signal performance criteria provide a means of evaluating the response of systems for incremental load changes, incremental voltage changes, and the incremental changes in synchronous machine rotor speed associated with the initial stages of oscillatory instability (where oscillations are small enough that nonlinearities are insignificant). Small signal performance data provide a means for determining or verifying excitation system model parameters for system studies (see IEEE Std 421.5, Hurley and Baldwin [B4], and Dandeno, et al. [B41]). The assumption of linearity limits the application of small signal models as noted above.

4.4 Effects of excitation limiters

A modern excitation system used on a synchronous machine may have sufficient excess capability to supply excitation under all operating conditions. The voltage regulator utilizes various auxiliary control and limiting functions that are designed to enhance power system performance and reliability while protecting both the excitation system and the synchronous machine. These limiting functions generally have no effect on the excitation output during normal operating conditions, but protect the exciter and machine during severe unit or power system disturbances in which the excitation system is pushed near or beyond defined operating limits (see Carleton, Bobo, and Burt [B40], Eberly and Schaefer [B42], Overexcitation Limiting Devices [B43], Underexcitation Limiter Models [B44], Nagy [B46], Ribeiro [B47], Rubenstein and Temoshok [B48]).

Typical excitation limiters include overvoltage, volts/hertz, overexcitation (both instantaneous and timed), underexcitation, and stator current limiters. These limiters may act to either decrease or increase the excitation level of the synchronous machine, depending on the type of limiter and the power system circumstances. Where a power system stabilizer (PSS) is applied, limiters should coordinate with PSS functions, and preferably not block the stabilizing signal. (This is more critical for underexcitation limiters as instabilities develop more readily at reduced levels of machine field current.)

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5. Large signal performance criteria

5.1 General

The following large signal performance criteria relate to excitation control systems and, where applicable, their components.

To permit maximum flexibility in the design, manufacture, and application of excitation systems, some of the performance criteria are defined under "specified conditions." The applicable conditions may be specified by the manufacturer or, more usually, by the equipment purchaser to meet applicable reliability or interconnection requirements. Care must be exercised that the "under specified conditions" clauses are interpreted in a manner consistent with the application.

It should be noted that some of the criteria discussed below cannot be directly tested and, thus, might depend on a combination of testing and simulation techniques to be verified.

When specified in per unit (pu), the base value (1 pu) of synchronous machine field current is normally considered to be the current required to produce rated terminal voltage on the air-gap line (see the annex in IEEE Std 421.5 on per unit system). The corresponding value of synchronous machine base field voltage is the air-gap field voltage as defined in Clause 3 and IEEE Std 421.1.

Sometimes, the requirements for ceiling current and ceiling voltage may be specified in multiples of the values that produce rated terminal voltage on the synchronous machine open-circuit saturation curve, or in percent above the values required when the synchronous machine is loaded at rated conditions. These values should not be confused with the defined bases for field current and voltage; hence specification of ceiling current and ceiling voltage in this way is discouraged.

When performing simulations to match dynamic responses obtained with the synchronous machine field at temperatures other than the values given in the definitions in Clause 3, it may be necessary to use different values of base field voltage to get the correct excitation system pu gains. However, all performance measures are based on the use of the specified field winding temperatures. Some simulations may include a field winding temperature (resistance) factor that can be adjusted to match responses obtained under various conditions.

5.2 Ceiling current

For some systems, the ability to attain ceiling current is important when the power system is operating at reduced voltage levels or under fault conditions to achieve maximum benefit to power system transient stability. In these cases ceiling current requirements with reduced excitation system supply voltage may need to be specified.

When sustained overloads or prolonged disturbances are of concern, the ceiling current may be based upon the excitation system thermal duty that identifies the maximum output current and the required time duration.

For some applications, the ceiling current will be determined by the requirement that the synchronous machine produces a specific value of steady-state, three-phase, short-circuit armature current for a specified time.

In some excitation systems (especially those that require a high ceiling voltage for forcing action), exciter current limiter control functions may determine the actual ceiling current. There may be separate requirements for short-term, mid-term, and sustained ceiling current for some excitation systems. The

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overexcitation limiter (OEL) and exciter or generator field overcurrent protection should be coordinated with these requirements (see IEEE Std C50.13™).

Also, in some special applications, the excitation system may be required to supply both positive and negative field current to the synchronous machine. Negative field current can be used in a salient pole synchronous machine, operating as a condenser, to increase the reactive power absorption. This can be useful near high-voltage direct current (HVDC) stations or long transmission lines where extra reactive power absorption is required. A dedicated thyristor bridge connected anti-parallel to the positive field current bridge is necessary to supply the negative field current. The negative field current ceiling requirement will normally be smaller than the positive ceiling value.

5.3 Ceiling voltage

The ceiling voltage provides an indication of the voltage available to force the field current from rated field current toward ceiling current. A greater difference between ceiling voltage and rated field voltage indicates greater forcing capability. Higher ceiling voltages tend to improve power system transient stability.

The no-load ceiling voltage is determined with the excitation system open-circuited (or with a high impedance load so minimal current will flow).

The excitation system ceiling voltage under load may be determined, as described in 8.2, by either a steady-state or transient measurement. The ceiling voltage is determined considering a specified supply voltage (usually rated voltage) and (if applicable) synchronous machine current.

For potential source and compound source excitation systems, whose supply depends on the synchronous machine voltage and current, the nature of power system disturbances and specific design parameters of the excitation control system influence the excitation system output. For such systems, the ceiling voltage is defined based upon a specified supply voltage and (if applicable) current. If testing is performed with a different supply voltage and (if applicable) current, analytical means may be used to predict the ceiling voltage with the specified supply.

Some excitation systems will have both positive and negative values of ceiling voltage. Positive ceiling voltage provides an indication of the voltage available to force the field current from rated field current toward ceiling current, while negative ceiling voltage provides an indication of the voltage available to force the field current to zero. Thus, large negative ceiling voltage aids in rapidly demagnetizing the synchronous machine.

For excitation systems employing a rotating exciter, the ceiling voltage is determined at rated speed.

5.4 Excitation system voltage response time

The excitation system voltage response time is the primary means for determining if an excitation system can be classified as high initial response. The voltage response time under load can only be determined by a transient measurement. The time measurement begins at the time the disturbance is initiated and ends when the dynamic response reaches 95% of the difference between ceiling voltage and rated field voltage. Adequate inductance is required in the load so that the time to reach 95% of the difference between ceiling voltage and rated voltage is measurable before the current begins to increase significantly above rated current and before any current limiters operate. The required load characteristics may not be practical to achieve in factory tests.

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5.5 High initial response (HIR)

For excitation systems with very short response time relative to the generator field time constant and to power system characteristic swings, the shape of the initial response is not of concern. Those systems whose voltage response time is 0.1 second or less are termed *high initial response (HIR)* excitation systems. The large signal response characteristics of these systems can be completely described by the ceiling voltage, voltage response time, and ceiling current.

5.6 Excitation system nominal response

The nominal response is used as a figure of merit for non-HIR excitation systems to permit comparisons of excitation systems. It is related to the ability of an excitation system to respond to transient stability events. The basic underlying concept of nominal response, as related to power system stability, is developed in Annex B. The half second interval was selected to associate the nominal response with the excitation system contribution to power system transient stability.

The starting point for determining the nominal response is the time at which the disturbance is initiated; that is, the excitation system nominal response should include any delay time that may be present. Referring to Figure 2, the excitation system nominal response is illustrated by line ac. This line is determined by establishing area acd = area abd. Typical values of nominal response range from 0.2 to 2.0 pu/s with higher numbers representing more contribution to power system transient stability.

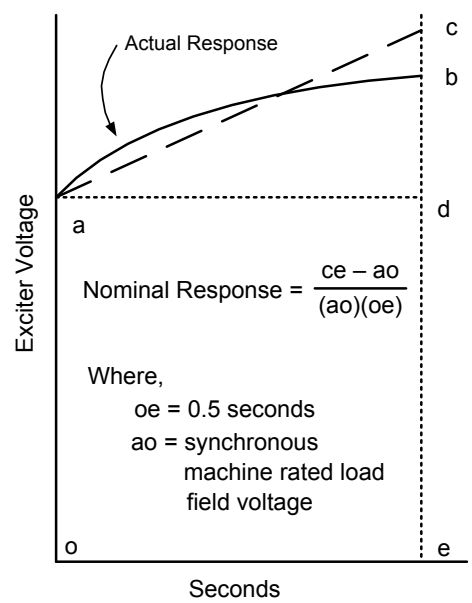

Figure 2—Excitation system nominal response

Nominal response is determined by initially operating the excitation system at rated field current of the synchronous machine and then suddenly creating the three-phase terminal voltage input signal conditions necessary to drive the excitation system voltage to ceiling (see IEEE Std 421.1). If nominal response is used as a figure of merit for comparing different types of excitation systems, misleading results may be obtained if different types of limiters or differing values of inductance are permitted. Nominal response determinations from actual dynamic responses obtained by recording responses during disturbances may be misleading when compared with factory tests because of the influence of fault induced currents in the generator field winding.

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5.7 Large signal dynamic response

Many of the large signal performance criteria provide a single numerical value. Excitation system ceiling voltage and nominal response are two examples. These criteria are useful for comparing performance of different systems and they are valuable as design criteria that can be used to verify that the manufacturer fulfilled specification requirements.

However, these single value criteria do not provide sufficient information in all instances for model parameter selection or verification. A match of these values can be achieved using models with different combinations of model parameters. Comparing model results with dynamic responses can reduce this ambiguity. Thus, dynamic response, also associated with small signal analysis, is a useful tool for large signal performance evaluation, especially for applications involving the refinement or validation of computer models.

A large signal dynamic response is a time response with the input and output variables of the component being tested, recorded as a function of time. Although dynamic response is generally associated with a step change in the input variable, it is only necessary that the change in the input variable be large enough and fast enough that the response at the output is classified as a large signal response for the result to be useful. The change to the input variable must be specified. Any portion of the excitation control system that is commonly represented in model studies (e.g., synchronous machine regulator, exciter, synchronous machine, excitation system, and excitation control system) may be evaluated using the dynamic response criteria. A dynamic response with several variables recorded simultaneously has considerable value in the refinement or validation of large signal computer models.

6. Small signal performance criteria

6.1 General

While the large signal performance criteria can be applied to the overall excitation control system, the excitation system, or individual components of the system, the small signal performance criteria are commonly used to evaluate the performance of closed-loop excitation control systems.

6.2 Small signal dynamic response

A typical dynamic response (to a step input) of a feedback control system is shown in Figure 3. The principal characteristics of interest are the 10% to 90% rise time, overshoot, peak time, and settling time as indicated.

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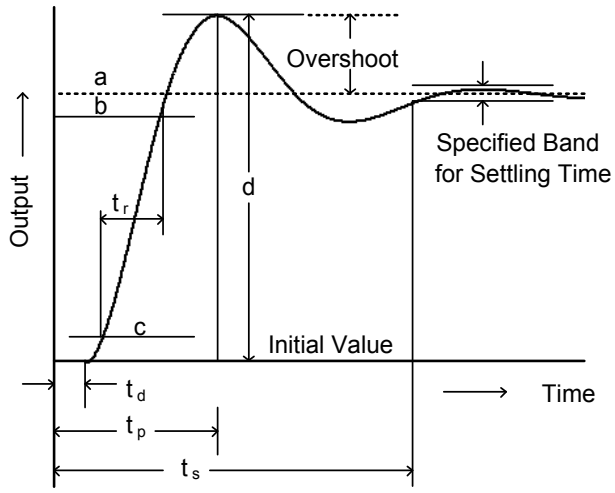

a: Steady-state value
b: 90% of steady-state value
c: 10% of steady-state value
d: Peak value

t_d: Delay time
t_p: Time to reach peak value
t_s: Settling time
t_r: Rise time

Figure 3—Typical dynamic response of a feedback control system to a step change in input

6.3 Frequency response characteristic (linear system)

In a linear system, the frequency dependent relation, in both gain and phase difference, between steady-state sinusoidal inputs and the resultant steady-state sinusoidal outputs is called a frequency response. Both open-loop and closed-loop frequency responses are useful for assessing the performance of feedback control systems. For excitation systems, frequency responses of the automatic voltage regulator control loop are of primary interest.

6.3.1 Open-loop frequency response

Typical open-loop frequency response characteristics of an excitation control system with the synchronous machine open-circuited (off-line) are shown in Figure 4. The principal characteristics of interest are the low frequency gain, G, gain crossover frequency, ω_c, phase margin, ϕ_m, and gain margin, G_m.

It is usually not practical to measure the open-loop frequency response characteristics of excitation control systems directly during field testing; however, many analysis techniques and instruments provide the tools for developing the open-loop transfer function of an excitation system from closed-loop measurements.

Open-loop frequency response characteristics are useful in determining gain margin and phase margin, both of which are measures of relative stability. Relative stability of a closed-loop control system can be determined from the properties of the Bode plot of the open-loop transfer function. However, this method can be used only if the open-loop transfer function does not have any poles and/or zeros in the right half s plane (minimum phase characteristics).

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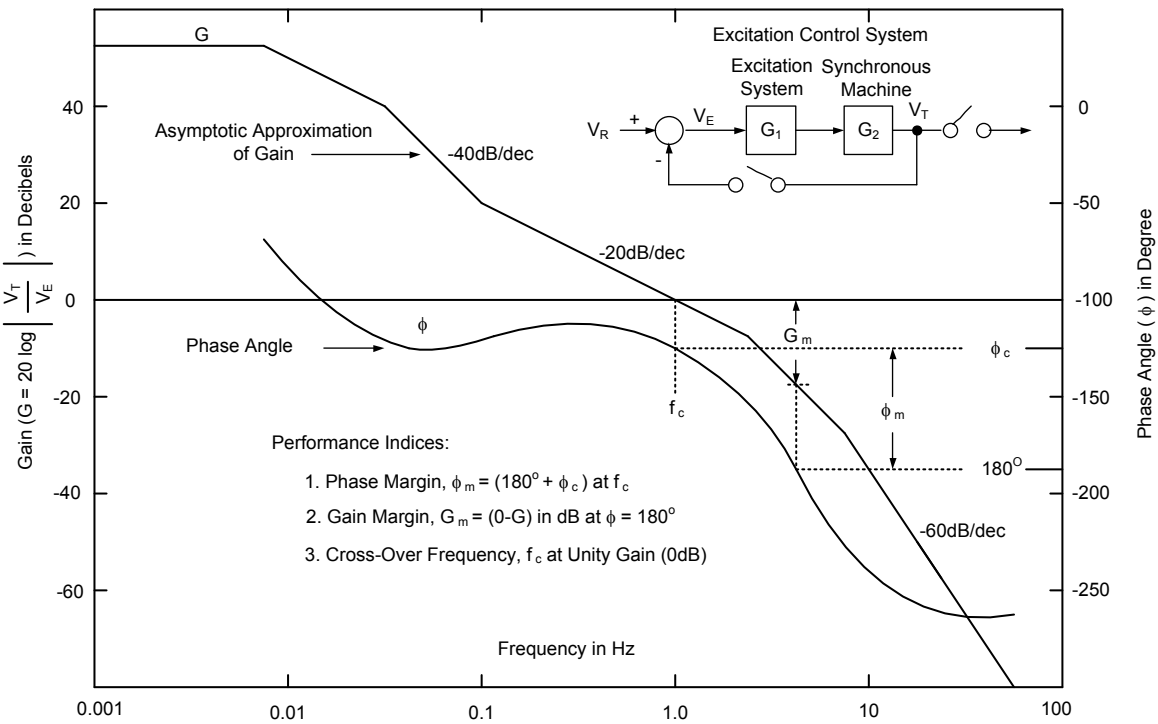

Figure 4—Typical open-loop frequency response of an excitation control system with the synchronous machine open-circuited

Relative stability of a feedback control system is measured in terms of the gain and phase margins. For a minimum phase system, which is stable with the feedback loop open, the gain and phase margins must be positive in order for the system to be stable with the feedback loop closed. Negative gain and phase margins mean that the system will be unstable with the feedback loop closed. In general, a gain margin of 6 dB or more and a phase margin of 40° or more is recommended for most feedback control systems.

The process of determining stability of non-minimum phase (poles and/or zeros in the right half s plane) and conditionally stable systems is relatively more involved and beyond the scope of this guide. A conditionally stable system is said to be stable for a given range of values of gain but becomes unstable if the gain is either reduced or increased sufficiently. Normally, conditionally stable excitation control systems should be avoided whenever possible.

6.3.2 Closed-loop frequency response

The corresponding closed-loop frequency response is shown in Figure 5. Here the parameters of interest are the bandwidth ω_B, the peak value M_p of the gain characteristic, and the frequency ω_m at which the peak value occurs.

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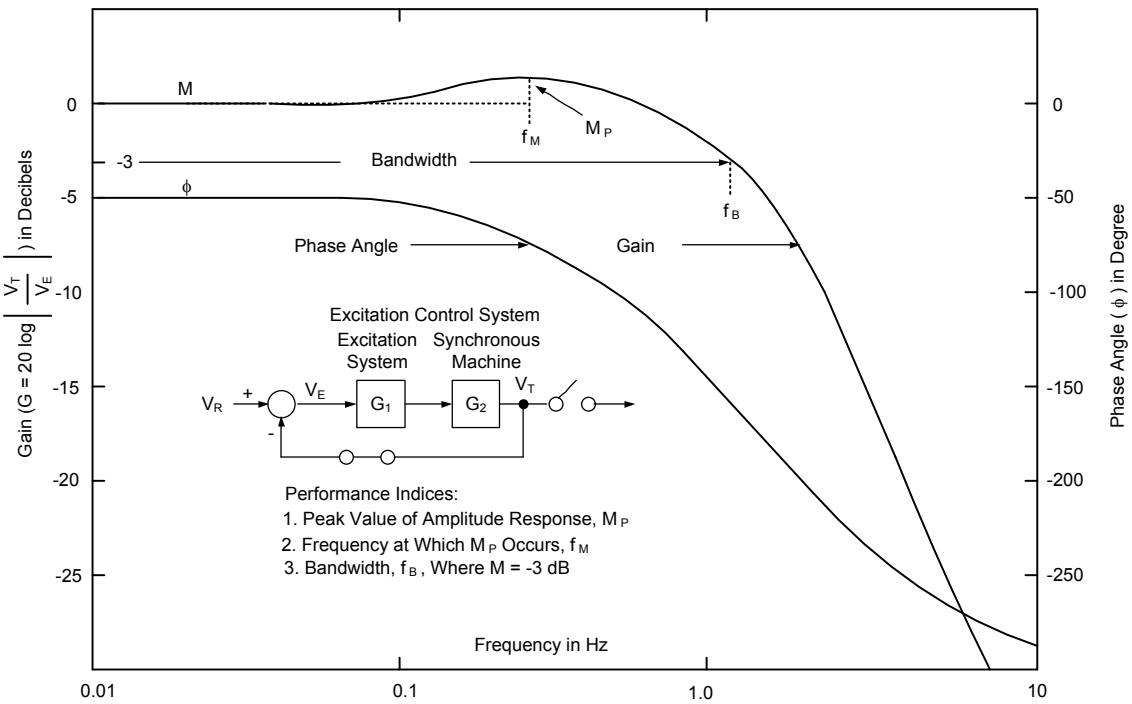

Figure 5—Typical closed-loop frequency response of an excitation control system with the synchronous machine open-circuited

With respect to the closed-loop frequency response characteristics, the peak value M_p ratio of the amplitude response is also a measure of relative stability. The peak value M_p is related to the damping of the closed-loop system. Analytical expressions relating M_p and damping are available at least for low-order systems and they show that increasing M_p reduces the damping of the closed-loop system. Thus, a high value of M_p (> 1.6) is indicative of an oscillatory system exhibiting large overshoot in its dynamic response. In general, $1.1 < M_p < 1.6$ is considered good design practice for most feedback control systems.

Bandwidth ω_B is a significant closed-loop frequency response performance index because it is indicative of the rise time T_r or speed of the dynamic response of the system; it measures, in part, the ability of the system to reproduce input signals. In feedback control systems having a step response exhibiting less than 10% overshoot, rise time T_r in seconds is related to bandwidth ω_B in rad/s by the approximate relationship:

$$T_r\omega_B = 0.30 \text{ to } 0.45 \quad (1)$$

In general, the product $T_r\omega_B$ increases as the overshoot in the system dynamic response increases; values in the range of 0.3 to 0.35 correspond to negligible overshoot; values in the region of 0.45 correspond to systems with about 10% overshoot. The effects of excitation system bandwidth as it relates to various types of synchronous machine oscillations are discussed in 7.1.

6.4 Complex frequency domain (s plane)

The dynamic characteristics of a control system can be represented by plotting the eigenvalues (or characteristic roots) of its transfer function in the complex frequency domain or s plane. Typical root locations of an excitation control system with the terminal voltage feedback loop open and the synchronous machine open-circuited are shown in Figure 6 (see also Larsen and Swann [B32]).

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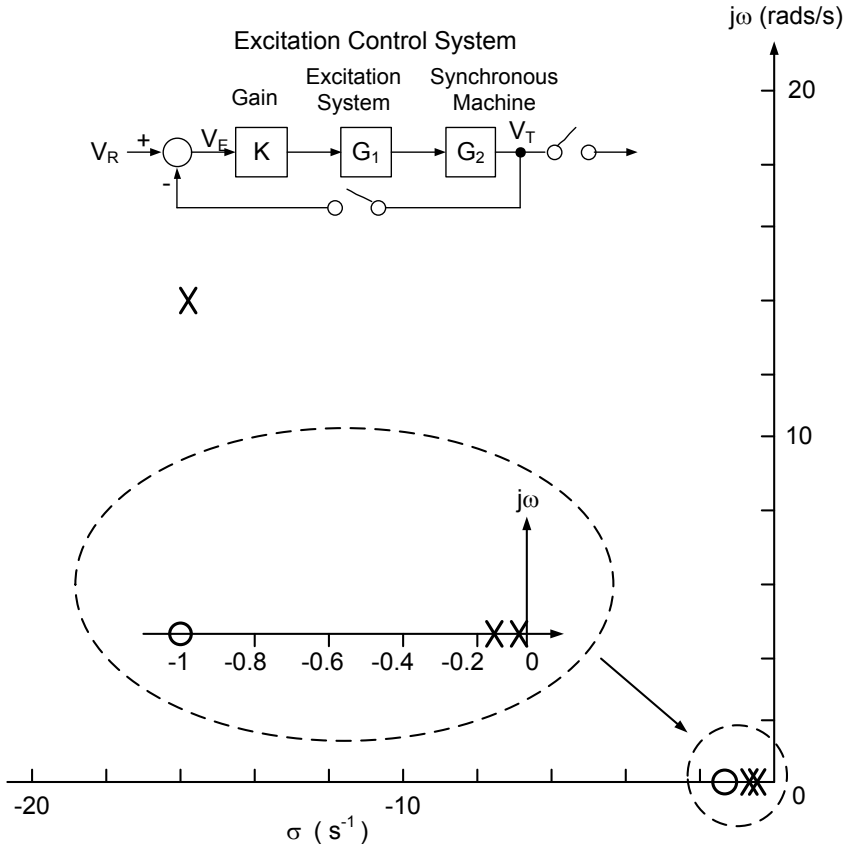

Figure 6—Pole/zero plot of a typical open-loop excitation control system with the synchronous machine open-circuited

Real poles and zeros (s = σ) are plotted on the horizontal axis of the s plane. Complex pole and zero pairs (s = σ ± jω) are typically plotted showing only the positive frequency root (σ + jω), with the negative frequency root (σ − jω) implied. The poles are the roots of the denominator of the transfer function (each indicated by "X"), while the zeros are the roots of the numerator (each indicated by "0").

Poles that are farther to the left of the jω (vertical) axis represent modes of oscillation that are more quickly damped than those nearer the jω axis. Poles that are on or to the right of the jω axis represent unstable modes and, thus, indicate a control system that is unstable.

The locations of the open-loop poles and zeros in Figure 6 (showing a proportional control system) depend upon the dynamic characteristics of transfer functions G_1 and G_2. Although the loop gain K has no effect on the open-loop poles and zeros, it has a large effect on the closed-loop poles. (For simplicity in this example, to help with understanding of root locus concepts, a proportional-only controller is considered as was typical in older products. Systems including an integral control term have an infinite steady-state gain in open loop which results in a zero steady-state error in closed loop.)

A root locus plot typically maps the locations of the closed-loop s plane poles as the loop gain is varied from zero to infinity. Figure 7 shows a root locus plot of the excitation control system with the open-loop characteristics of Figure 6. The poles of the closed-loop system are plotted in the s plane as the value of gain K is varied. At a value of gain K = 0, the closed-loop poles are the same as the open-loop system poles. At a value of gain $K = K_{co}$, complex poles cross over the jω axis into the right half plane, indicating instability. If the gain K is adjustable, the operating gain $K = K_{op}$ can be selected for acceptable gain margin (G_m) and damping ratio (ζ).

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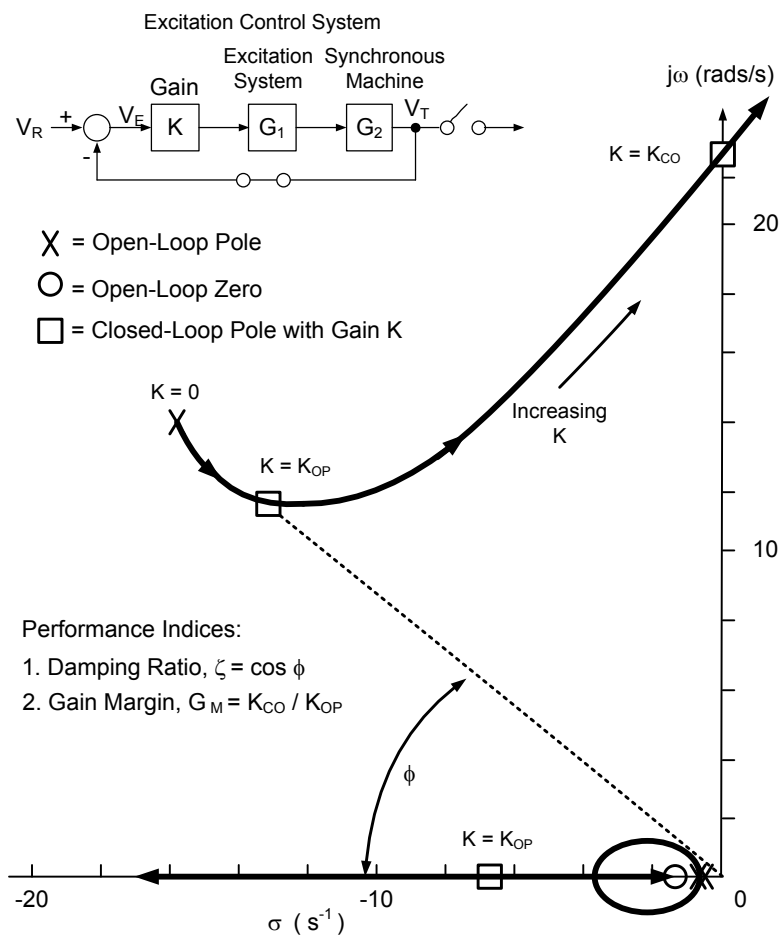

Figure 7—Root locus plot of a typical closed-loop excitation control system with the synchronous machine open-circuited as the gain (K) is incremented from zero to infinity

Generally accepted values of the performance indices described above, characterizing good feedback control system performance, are listed in Table 1.

Table 1—Generally accepted values of indices characterizing good feedback control system performance

Gain margin	≥ 6 dB
Phase margin	≥ 40°
Overshoot	0% to 15%
M_p	1.1 to 1.6 (0.83 dB to 4.0 dB)
Damping ratio	≥ 0.6

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6.5 Small signal performance indices

It is not possible to define such generally acceptable ranges of values for the other small signal performance indices: rise time, settling time, and bandwidth. These indices are measures of relative speed and stability of control action. In most feedback control systems, they are determined primarily by the dynamic characteristics of the system element whose output is the ultimately controlled variable. In the case of an excitation control system, the dynamic characteristics of the synchronous machine (field time constant, etc.) are the determining factors.

It should be noted that simultaneous optimization of all small signal performance indices is not possible. For example, low M_p, high damping ratio, high gain margin, and high phase margin are not compatible with maximum bandwidth. The performance indices which are of primary importance depend on the individual application of each feedback control system, and no universally applicable "best criteria" can be recommended in standards.

Typical ranges of values of small signal performance indices for an excitation control system are given in Table 2. These data have been derived analytically using the anticipated extreme ranges (longest to shortest) of synchronous machine field time constants and excitation system time constants likely to be encountered. Generally, values outside the ranges in Table 1 should be avoided by adjusting control parameters; however, some excitation systems may exhibit larger overshoot and smaller damping ratios, especially systems with rotating power amplifiers and/or rotating main exciters when the synchronous machine is disconnected from the power system. Therefore, some of the values listed in Table 2 are outside of the ranges of Table 1 in order to achieve the desired dynamic performance with the machine on-line. In these cases, overshoot and settling times may be at the higher end of the listed ranges when the machine is off-line.

Table 2—Range of excitation control system small signal dynamic performance indices

Performance index	Range of expected values
Excitation system loop gain K (for a proportional gain system as described in 6.4 above)	30 pu to 800 pu
Gain margin	2 dB to 20 dB
Phase margin	20° to 80°
M_p	1 to 4 (0 dB to 12 dB)
Bandwidth	0.3 Hz to 5 Hz
Overshoot	0% to 40%
Rise time	0.025 s to 2.5 s
Settling time	0.2 s to 10 s
Damping ratio	0.25 to 1
Voltage regulation accuracy (as defined in IEEE Std 421.1™)	0% to 1%

The performance indices in Table 1 are applicable to any feedback control system having a single major feedback loop, that is, a single ultimately controlled variable. As such, they are applicable to an excitation control system with the synchronous machine open-circuited. Figure 8 shows the terminal voltage behavior for a +/− 3% step response on the automatic voltage regulator (AVR) reference of synchronous machine with brushless exciter in off-line and on-line conditions. A loaded synchronous machine connected to a power system is a complex multi-loop, multi-variable feedback control system.

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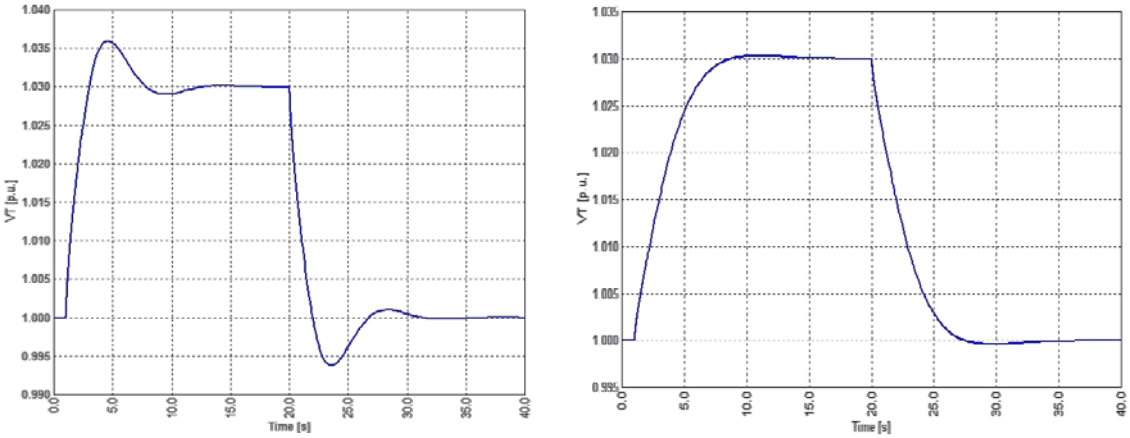

Off-line (generator open circuit) On-line (generator connected to power system)

Figure 8—Comparison of terminal voltage step responses for brushless machine

For a loaded synchronous machine connected to a multi-machine interconnected power system, performance indices such as those of Table 1 lose much of their significance. Analysis and synthesis techniques that are applicable to these types of systems should be used. One approach to assessing the performance of this complex system is to model the system using state space techniques and calculate the eigenvalues (characteristic equation roots) for the range of excitation system, synchronous machine, and system parameters of interest. The state space model can be derived from known plant and system parameters, operating conditions, and experimental frequency response data, if the latter is available. The calculation of eigenvalues gives a direct indication of system stability for the linearized system and serves as an efficient initial step before testing the selected parameters in a more extensive study such as a transient stability study. The quadratic performance index and other indices that utilize the state space models are measures of the ability of a multi-loop control system to meet specified criteria.

When a synchronous machine is connected to a power system, its operating level, and the parameters of the external system greatly influence performance. Figure 9 illustrates some of these effects showing calculated frequency response characteristics and s plane pole/zero locations of the transfer function $\Delta V_T/\Delta E_{FD}$ for a typical large synchronous machine at three operating points: torque angles of 0°, 72°, and 102°, respectively.

The characteristics of Figure 9 are calculated for a synchronous machine connected to an infinite bus through an external reactance with no synchronous machine regulator or governor action. Torque angle depends on machine characteristics, machine operating point, and on the magnitude of external impedance.

Figure 9 shows that the dynamic characteristics of a synchronous machine change dramatically when the torque angle increases. Figure 9(a) shows that an unloaded synchronous machine (δ = 0°) behaves as a simple inductive circuit with a single time constant, corresponding to the time constant of the field circuit. The maximum phase lag of 90° occurs when the frequency, ω, reaches infinity. In the s plane, the machine exhibits a complex pair of poles and zeros that cancel each other in the $\Delta VT/\Delta EFD$ transfer function.

Figure 9(b) and Figure 9(c) show frequency responses with characteristic resonant peaks appearing in the gain function accompanied by a sudden dip in the phase angle. These characteristics are attributable to poorly damped complex poles introduced into the transfer function of the synchronous machine by an increase in the torque angle. In the s plane plots of Figure 9(b) and Figure 9(c), the complex pair of zeros no longer coincides with the complex pair of poles. In Figure 8(c) (δ = 102°), the s plane pole/zero plot shows a pole on the positive real axis.

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This indicates monotonic instability, as would be expected with an uncontrolled synchronous machine operating at a torque angle greater than 90° (Larsen and Swann [B32]). The highly oscillatory characteristics of a synchronous machine operating at high torque angle can readily lead to instability when the machine is operated in a closed-loop feedback control system with a voltage regulator. Application of high initial response excitation systems with ceiling voltage capability and high closed-loop gain improves both transient and steady-state stability. However, these systems inherently reduce damping of small signal oscillatory modes.

It is frequently necessary to apply supplementary control, such as a power system stabilizer, to compensate for these characteristics to ensure stable operation and improve damping of oscillations arising from system disturbances.

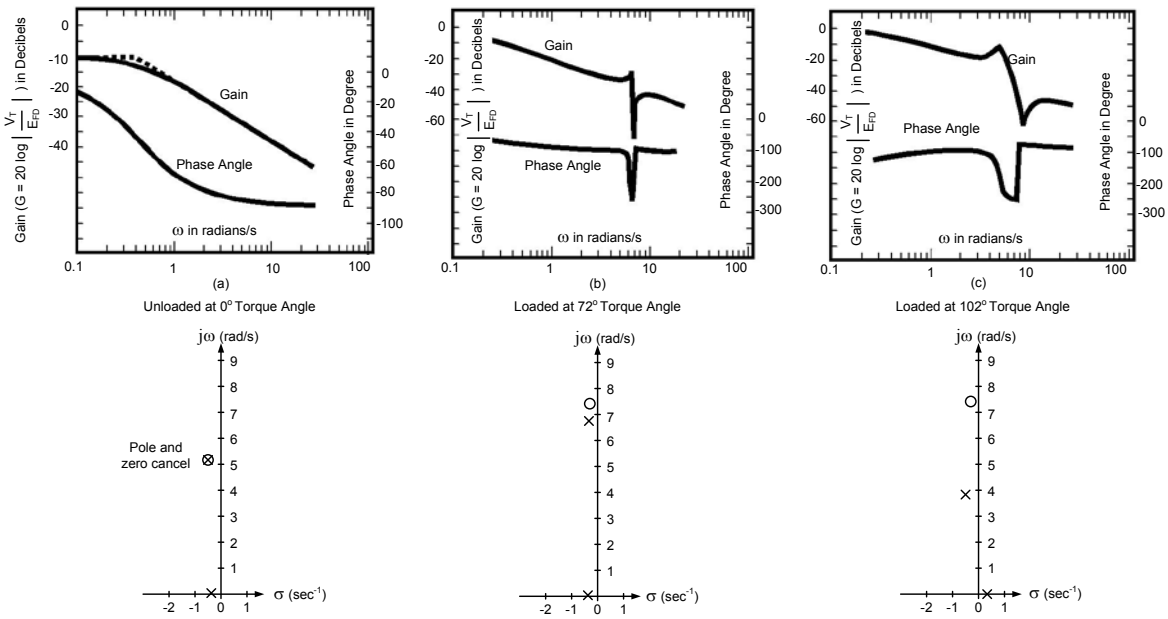

Figure 9—Frequency response characteristics and s plane pole/zero locations of a typical synchronous machine as a function of torque angle, with no voltage regulator action and with no speed governor

7. Excitation control system effects on power system stability

Many factors influence the response and performance of an interconnected generator, excitation system, and power system during a disturbance. Of these factors, it is well known that generator loading and excitation system parameter settings can greatly affect the oscillatory nature of the response (Byerly, Keay, and Skooglund [B13], DeMello and Concordia [B16], Kundur [B29]). Reduced stability margins occur due to increasing generator output which increases internal generator rotor angle, or due to increasing power across a transmission interface which increases transmission angles. The dynamics of system stability are also greatly influenced by excitation control. High AVR gain settings increase synchronizing torques, but at the same time decrease damping torques (Byerly, Keay, and Skooglund [B13], DeMello and Concordia [B16], Kundur, et al. [B30], Larsen and Swann [B32]).

The excitation control system response is a combination of many modes of oscillation, each with a different frequency, phase, and amplitude. The nature of each mode, whether it is a local, interarea or interunit mode, is determined by the operating condition of the excitation control system. Understanding the influence of

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the excitation system on these modes is important when designing the excitation system. The material in this section describes the different mode types and the application of power system stabilizers.

7.1 Types of synchronous machine oscillations

The oscillatory relationships between a synchronous machine and a power system, and among synchronous machines on a power system, should be carefully considered whenever applying excitation controls. If an excitation system in its particular application has the capability of destabilizing any particular mode of oscillation, then supplementary controls may be necessary to prevent or minimize the destabilizing effect. An excitation system may also be called upon to provide damping to oscillatory modes that may already exist on a power system.

Synchronous machine oscillations can often be characterized as falling into one of the four categories described below. Often, the frequency of the oscillation will provide the best indication of which type of oscillation is occurring. For this reason, power plant operators should be instructed to try to note the swing frequency or period of any significant oscillation whenever it is observed.

Of the four categories of oscillations described below it is important to note that excitation systems may include power system stabilizer algorithms to increase damping for the first three categories of oscillations, which can be described as electro-mechanical oscillations. However excitation system power system stabilizers should not be applied to reducing the fourth category, torsional oscillations, and it is important to ensure that stabilizers are programmed not to adversely respond to the torsional oscillation frequency range.

7.1.1 Local machine system oscillations

These oscillations generally involve one or more synchronous machines at a power station swinging together against a comparatively large power system or load center at a frequency in the range of 0.7 Hz to 3.0 Hz. These oscillations become particularly troublesome when the plant is at high load with a high reactance transmission system.

The use of modern excitation systems allows faster terminal voltage regulation than was possible with older, more sluggish excitation systems. Although the synchronizing torque coefficient is enhanced as the gain of the synchronous machine regulator is increased in the range of the oscillation frequency, negative damping is introduced into the machine torque speed loop (see DeMello and Concordia [B16], IEEE Tutorial Course [B23]). Depending on the desired dynamic gain and the system in which it is employed, local machine system oscillations may become negatively damped. To achieve high synchronizing torque, power system stabilizers may be used to restore the damping torque contribution of the exciter as described in 7.2.

A fast-acting or wide bandwidth excitation system has a greater potential for destabilizing local machine system oscillations. This often necessitates the application of a PSS on this type of excitation system. In some situations when the PSS is out of service the excitation system has to be operated at reduced gain or placed on manual control to prevent the high gain regulator setting from creating an unstable situation. For newer digital excitation systems, two AVR gain setting groups may be provided, one setting group for the PSS on and another setting group for the PSS off (Basler, et al. [B1]). When the PSS is turned off, a lower AVR gain is utilized. The application of a properly tuned power system stabilizer on most modern normal response or high response excitation systems will contribute positive damping to these types of oscillations. In this case, a wider bandwidth excitation system has the potential to contribute a greater amount of damping, although it has a greater reliance on proper power system stabilizer operation. Some excitation systems are equipped with controls to automatically reduce exciter gain, or transfer to an auxiliary channel when the stabilizer is out of service; other systems may have redundant stabilizers (Kundur, et al. [B31]).

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7.1.2 Interarea oscillations

These oscillations usually involve combinations of many synchronous machines on one part of a power system swinging against machines on another part of the system. Interarea oscillations are normally lower in frequency (< 1.0 Hz) than local oscillations. Therefore, older excitation systems even with a quite sluggish response, as well as more modern excitation systems, may contribute either positive or negative damping to interarea oscillations, depending upon the generator's location in the power system and whether or not a power system stabilizer is applied. Since interarea oscillations usually involve many machines, successful damping of such modes may require the application of power system stabilizers on the excitation systems of a large number of machines (Klein, Rogers, and Kundur [B26], Klein et al. [B27], Kundur [B29], Kundur et al. [B30]). An example of low frequency interarea oscillations following a major disturbance is illustrated in Figure 10. In this example the swing frequency is approximately 0.35 Hz and the damping ratio is about 0.2.

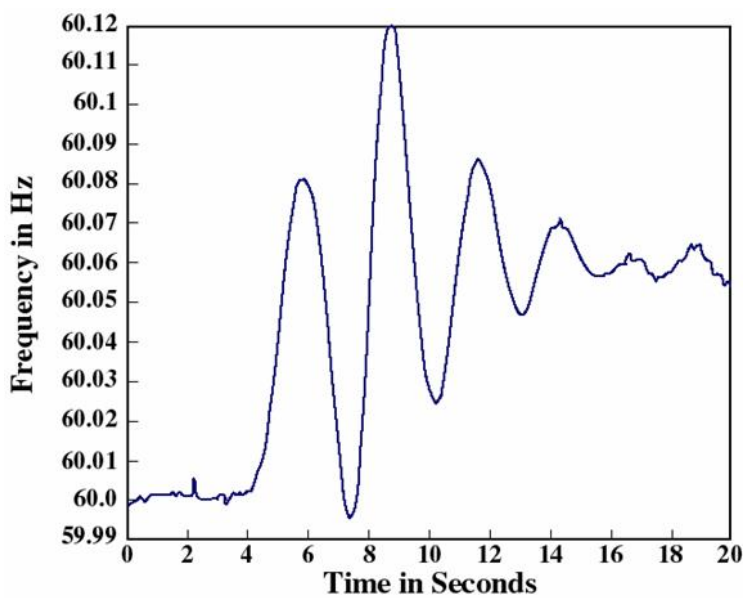

Figure 10—Power system interarea oscillation following a major disturbance

7.1.3 Interunit oscillations

These oscillations typically involve two or more synchronous machines at a power plant or nearby power plants in which machines swing against each other, usually at a frequency of between 1.5 Hz and 3 Hz. These oscillations can be destabilized or stabilized with a power system stabilizer, and thus may require consideration in determining power system stabilizer settings.

7.1.4 Torsional oscillations

These oscillations involve relative angular motion between the rotating elements (synchronous machine, turbine stages, and rotating exciter mounted on the same shaft) of a unit (usually a steam-driven system), with frequencies higher than electro-mechanical modes e.g., typically greater than 10 Hz for 3600 rpm turbines or as low as 5 Hz for 1800 rpm turbines. The turbine-generator mechanical system has very little inherent damping. The source of torque for inducing torsional oscillations with the excitation system comes from modulation of synchronous machine power due to changes in generator field voltage. Besides the excitation systems, there are other mechanisms which can excite torsional oscillations such as unbalanced

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faults, HVDC converter control, static converter control, governor control, and transmission series capacitors (Lawson, Swann, and Wright [B45], "Third Supplement" [B49], "Countermeasures" [B50]).

The ability of the excitation system to excite torsional oscillations is enhanced in wider bandwidth excitation systems. A wide bandwidth excitation system may have the capability to provide enough negative damping at the lower torsional natural frequencies to destabilize one or more of these torsional modes, particularly with the application of a power system stabilizer deriving input from speed deviation alone. If the excitation system has sufficient gain at the torsional frequency of interest, oscillations may quickly build to a level resulting in damaging stresses in the shafts. Watson and Coultes [B38] vividly illustrates this problem. The problem is compounded by the fact that power plant operators may be unaware that unstable torsional oscillations are occurring, because typical recording and indicating meters do not respond to these frequencies. Figure 11 shows data recorded during excitation system commissioning.

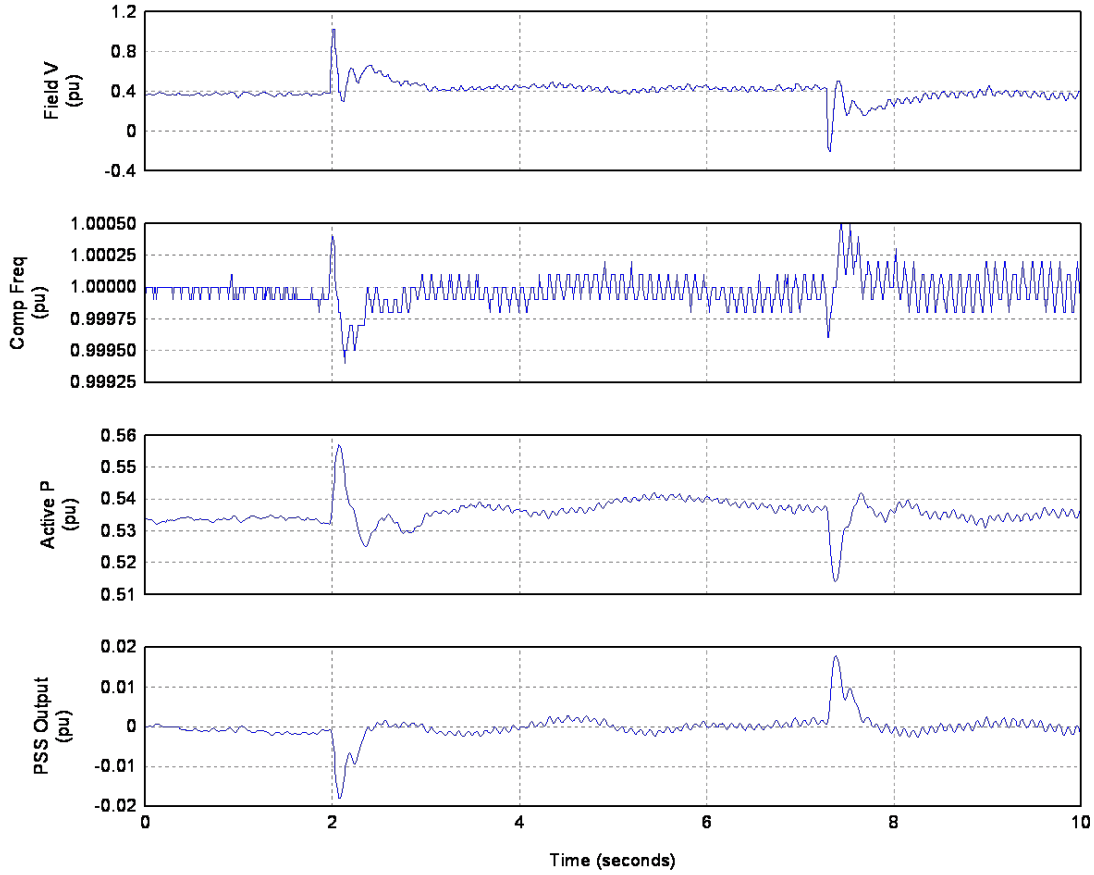

Figure 11—Torsional oscillations recorded during excitation system commissioning

Analysis of the effects of excitation control on torsional oscillations generally requires more complex excitation system and synchronous machine multi-mass models than those necessary for analyzing local machine system oscillations and other types of power system/synchronous machine oscillations. Ultimately, tests using instruments capable of responding to torsional frequencies and measuring speed sensors may be necessary to confirm that torsional modes will not be destabilized, or that countermeasures such as notch filters in the power system stabilizer, or choice of multi-input PSS design such as the PSS2B or PSS4B, are acting to prevent these oscillations.

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From the series compensated (SC) network example (Second Benchmark Model [B51]) in EMTP-RV, Figure 12 below illustrates torsional modes observable on the electrical power and the current of the machine after a fault and loss of one line. Note the following key points:

— The SC zero impedance is tuned at 22 Hz and appears on machine currents at 22 Hz and on Pe and PSS output at 38 (60-22) Hz.

— From the ten mass data, the four most important torsional modes are tuned at 10.4, 18, 54, and 56 Hz and appear on the mechanical torques for the lower mode on Pe and the PSS output.

— Existing torsional mode problems may be exacerbated when the output of the PSS includes frequencies which coincide with frequencies of torsional modes. In the case of this example they do not coincide.

In all cases the PSS should avoid amplifying these frequencies.

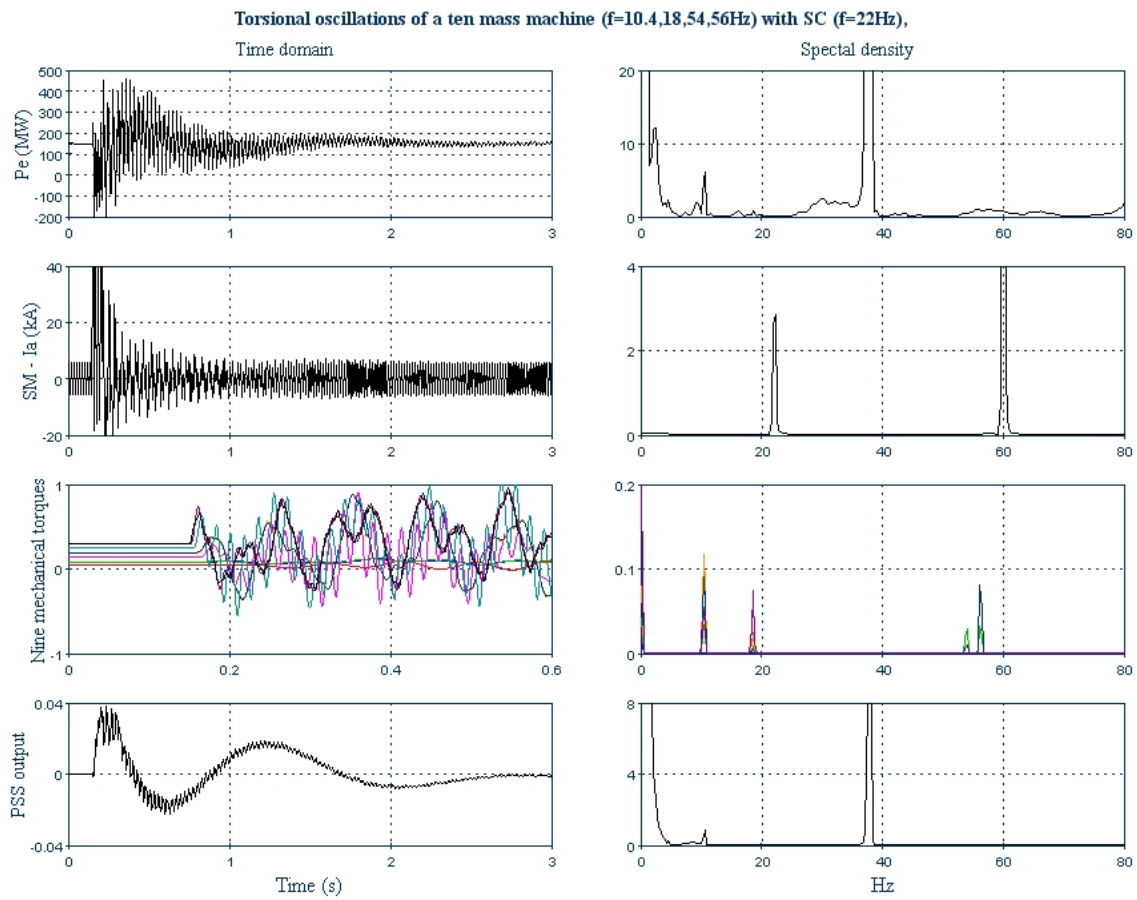

Figure 12—Example of torsional oscillations from simulations

7.2 Application of power system stabilizers

By the mid-1960's several authors had reported successful experience with the addition of supplementary feedback to enhance damping of rotor oscillations (Dandeno, et al. [B15], Schleif [B35]). Some of the earliest work on modeling of excitation systems and their role in rotor-angle stability still provides a good starting point for understanding the topic (see DeMello and Concordia [B16] and Heffron and Phillips

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[B22]). Readers seeking information on modern techniques applied to simulating and tuning these systems for the improvement of overall system stability are directed to reference (Kundur, et al. [B30]). IEEE Tutorial Course [B23] and Larsen and Swann [B32] provide some guidance on test techniques that may be used to assess the performance of power system stabilizers on operating generators. Textbooks such as Byerly and Kimbark [B14] and Kundur [B29] provide in-depth explanation of power system stability phenomena and generator control modeling.

A power system stabilizer is a control function that provides an additional input to the automatic voltage regulator to improve power system dynamic performance. The input to the power system stabilizer may be shaft speed, frequency, synchronous machine electrical power, accelerating power, integral of accelerating power, or a combination of these. These electrical quantities are typically derived from machine voltage and current signals through appropriate transducers that are an integral part of the power system stabilizer.

Transfer functions of typical single-input, dual input, and multi-band and example settings for power system stabilizers are shown in IEEE Std 421.5. The parameter values depend upon the nature of the input signal and the specific application of the stabilizer. Design ranges of typical parameters for power system stabilizers that have been applied in North America are given in Table 3, Table 4, Table 5, and Table 6. These tables are provided for information only should not be considered as representative of all possible successful designs.

Regardless of the source for the power system stabilizer control signal, the control signal transducer should not introduce extraneous signals within the control system bandwidth. Without the control signal connected to the transducer input, the output of the power system stabilizer following the signal conditioning and reset stages should be relatively free from noise generated by the transducer or signal conditioning stages. With the signal conditioning circuitry adjusted as in actual operation, the total influence of all noise sources should not exceed 1% of the dynamic range of the output of the power system stabilizer, unless otherwise specified.

The fundamental function of the signal-conditioning network of the power system stabilizer is to compensate for the phase lags of the system being controlled. Phase compensation is generally accomplished by the use of lead-lag functions providing phase compensation over the frequency range of interest. The phase compensation requirements are largely controlled by the closed-loop voltage regulator response. There is considerable variation in these responses and, as a result, PSS phase compensation is commonly implemented using one to three stages of phase lead providing up to 150 degrees of phase advance.

Table 3—Range of typical design parameters for power system stabilizers with frequency or speed input (Type PSS1A)

Symbol	Typical range	Definition
T_6	0 s to 0.04 s	Transducer time constant
T_5	0.5 s to 50 s	Washout (reset) time constant
T_1, T_3	0.03 s to 2.0 s	Lead (zero) time constant
T_2, T_4	0.01 s to 10 s	Lag (pole) time constant
K_S	0.10 pu to 10 pu	Stabilizer gain
V_{STMIN}, V_{STMAX}	±0.02 pu to ±0.10 pu	Stabilizer output signal limits

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Table 4—Range of typical design parameters for power system stabilizers with power input (Type PSS1A)

Symbol	Typical range	Definition
T_6	0 s to 0.04 s	Transducer time constant
T_5	0.5 s to 50 s	Washout (reset) time constant
T_1, T_3	0.1 s to 2.0 s	Lead (zero) time constant
T_2, T_4	0.01 s to 0.20 s or 10 s to 20 s	Lag (pole) time constant
K_S	±0.10 pu to 10 pu	Stabilizer gain (sign depends upon choice of input signal)
V_{STMIN}, V_{STMAX}	±0.02 pu to ±0.10 pu	Stabilizer output signal limits

Table 5—Range of typical design parameters for dual input power system stabilizers (Type PSS2A/B)

Symbol	Typical range	Definition
T_6	0 s to 0.04 s	Transducer time constant
T_{W1} through T_{W4}	0 s to 20 s	Washout (reset) time constants[a]
K_{S2}	0 pu to 10 pu	Mixing gain
K_{S3}	1	Mixing gain[b]
T_8, T_9, N, M	T_8 = 0.5 s; T_9 = 0.1 s; N = 1; M = 5 or T_8 = 0.3; T_9 = 0.15; N = 4; M = 2	Selected to minimize voltage change during mechanical power changes[c]
T_7	0 s to 20 s	Low pass filter time constant[b]
T_1, T_3, T_{10}	0.01 s to 6.0 s	Lead (zero) time constant
T_2, T_4, T_{11}	0.01 s to 6.0 s	Lag (pole) time constant
K_{S1}	0.10 pu to 50 pu	Stabilizer gain[d]
V_{STMIN}, V_{STMAX}	±0.02 pu to ±0.10 pu	Stabilizer output signal limits[d]

[a] A value of 0 indicates a bypassed block.
[b] When the PSS2A or PSS2B structure is used to represent integral-of-accelerating-power-based PSS units K_{S3} = 1, T_7 = T_{W2}, T_{W4} = 0, K_{S2} = T_7/(2 × inertia).
[c] Some special circumstances may require alternative selection of T_8, T_9, N, and M.
[d] V_{STMAX} and K_{S1} typical values assume V_{REF} summation point PSS.

Table 6—Range of typical design parameters for multi-band power system stabilizers (Type PSS4B)

Symbol	Typical range	Definition
F_L	0 Hz to 0.3 Hz	Low band central frequency
F_I	0.3 Hz to 2 Hz	Intermediate central frequency
F_H	2 Hz to 10 Hz	High band central frequency
K_L	0 pu to 10 pu	Low band gain
K_I	0 pu to 50 pu	Intermediate band gain
K_H	0 pu to 100 pu	High band gain
VL_{MIN}, VL_{MAX}	0 pu to 1.0 pu	Low band output signal limits
VI_{MIN}, VI_{MAX}	0 to 1.0 pu	Intermediate band output signal limits
VH_{MIN}, VH_{MAX}	0 to 1.0 pu	High band output signal limits
V_{STMIN}, V_{STMAX}	0 to 0.1 pu	Stabilizer output signal limits

NOTE—The settings of the PSS are selected from the six parameters F_L, F_I, F_H, K_L, K_I, K_H

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$$T_{L2} = T_{L7} = \frac{1}{2\pi F_L \sqrt{R}}$$

$T_{L1} = T_{L2}/R$

$T_{L8} = T_{L7}\, 3R$

$K_{L1} = K_{L2} = (R^2 + R)/(R^2 - 2R + 1)$

R is a constant equal to 1.2 in this example, and all other time constants are set to 0.

For the best contribution to system damping, the gain K_S of the power system stabilizer should be maximized within the constraints imposed by stability of the power system stabilizer control loop or any other synchronous machine modes of oscillation as discussed in 9.4.

The speed change $\Delta\omega$ derived from any of the stabilizer inputs is a change with respect to a fixed frequency reference. The desired stabilizing signal has as its reference the base system frequency. This system frequency is not fixed but is changing (normally slowly) all the time. This change causes the terminal voltage to vary. A washout function (high-pass filter) serves to minimize the effect of overall system speed changes on machine terminal voltage. Choice of washout time constant is not critical except that:

a) It should be long enough so that its phase shift does not interfere significantly with the signal conditioning network at the desired frequencies of stabilization, especially at low-frequency interarea modes.

b) It should be short enough so that the terminal voltage will not be unduly affected by overall system speed variations, considering system islanding conditions where applicable. In some cases utilities have elected to set washout time constants from 10 s to 15 s even in islanding scenarios.

When synchronous machine electrical power is used as an input to the PSS, operation during mechanical power changes must be considered. When mechanical (and electrical) power changes, the stabilizer output and terminal voltage reference will also change, unless the PSS structure or settings are selected carefully. With previous designs of power-based stabilizers, high-pass (washout) time constants would be limited below 5 s to help reset mechanically induced PSS output changes. With some stabilizer structures (e.g., integral of accelerating power with ramp tracking mechanical power filter) the washout time constants are not limited, and high rates of mechanical power change are acceptable without producing objectionable terminal voltage variations.

Most power system stabilizers incorporate limits on the output signal. Limiting may be accomplished by simple clipping of the stabilizer output signal as shown in IEEE Std 421.5 (stabilizer parameters V_{STMIN} and V_{STMAX}) or it may be accomplished by control action through a terminal voltage limiter (some PSS designs include an internal terminal voltage limiter function for this purpose). In a few applications, limiting has been accomplished by disconnecting the stabilizer when terminal voltage deviation reaches preselected values; however, this practice can lead to limit cycling and other undesirable effects.

The high-frequency filters are included as a means of representing the phase-shifting effects of shaft torsional filters (which are used with some power system stabilizers) as well as other high-frequency time constants within the power system stabilizer used to reduce high-frequency gain. They should not significantly detract from the primary control function of the power system stabilizer, although they may add an additional phase lag to the lower frequency modes that must be compensated by additional phase lead in the lead/lag stages. This should make these transfer functions more closely represent the actual equipment.

Analytical studies and field test results of applications of power system stabilizers in North America are contained in references [B11], [B12], [B13], [B16], [B18], [B19], [B20], [B21], [B23], [B25], [B28], [B31], [B32], [B34], [B35], [B36], [B37], [B38], [B39]. Suggested procedures for field testing and tuning

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of power system stabilizers are given in 9.3 (see also IEEE Tutorial Course [B23]). A block-diagram approach was first used by Heffron and Phillips [B22], and later by DeMello and Concordia [B16], to study the steady-state stability characteristics of synchronous machines. Kundur [B29] provides additional detail. The approach to AVR and PSS tuning documented in Kundur, et al. [B30], can produce the greatest improvement in both synchronizing and damping torques through excitation control.

Extensive information on the application of power system stabilizers can be found in IEEE Std 421.5™ and IEEE Tutorial Course [B23].

8. Large signal performance testing

8.1 General

Standard procedures for large signal performance testing are not possible for all applications. Test procedures performed on small units will not have the same impact on the power system that the same procedures might have on a large unit. The impact of testing is determined to a large extent by the nature of the power system surrounding the unit being tested. For many units, these considerations extend to the auxiliary equipment electrically connected to the unit under test.

For these reasons, large signal testing is not always practical under specified conditions. In such cases, model results and/or calculations using applicable test data can be used to predict the performance under the specified conditions.

Large signal performance tests are used to demonstrate that the manufacturer has met the specified requirements of the customer (acceptance testing), to determine parameters for models used to predict large signal performance, and to ensure that the equipment is working correctly (periodic maintenance testing). When large signal tests are a part of acceptance test procedures, the criteria to be used in evaluating test results must be specified. The user and manufacturer must agree on which tests are to be performed, which tests are to be performed at the factory, which tests are to be performed at the installation site, and which tests can be replaced by results from model studies and/or calculations. Tests for model verification require close attention to specification of the conditions under which the tests are performed and/or close attention to the power system conditions used in the model.

Excitation systems may have asymmetrical large signal responses due to different positive and negative forcing characteristics. The test methods and performance criteria for decreases in excitation are similar to those for increases in excitation.

8.2 Factory testing

Most of the large signal tests were developed to permit acceptance testing to determine that the manufacturer has fulfilled the requirements of the customer's specifications. Ceiling voltage, ceiling current, voltage response time, and nominal response are examples. The manufacturer uses these large signal performance criteria to determine that the equipment meets the user's specifications. Factory testing is typically performed only on prototypes of new equipment.

Factory ceiling voltage tests under load on larger static excitation systems are often impractical to perform. In these cases, analytic determination based on measured data of the excitation transformer (secondary voltage, impedance), cable/bus duct connections (impedance), and power converter (voltage drop,

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minimum and maximum firing angle) may be made or tests may be performed in the field at the installation site.

The excitation system ceiling voltage under load may be determined by either a steady-state or transient measurement. The ceiling voltage is determined considering a specified supply voltage (usually rated voltage) and (if applicable) synchronous machine current.

Steady-state determination of ceiling voltage may be performed by loading the excitation system initially with a high resistance then reducing the resistance such that rated current is attained, but not exceeded, with the excitation system attempting to operate at ceiling voltage output. It is necessary that the load has adequate inductance so that current and voltage waveforms are reasonably duplicated. The steady-state ceiling voltage measurement is made with the exciter output current limiter control circuits inactive.

For transient determination of ceiling voltage adequate inductance is required in the load so that the ceiling voltage is measurable at rated current before any current limiters operate. Because of commutation voltage losses in a power semiconductor bridge, the available ceiling voltage is reduced as current increases. For this reason it is important to measure the ceiling voltage of HIR exciters a very short time (50 ms) after the event is initiated so that current does not change appreciably.

8.3 Field testing

The procedures for any field test to determine large signal performance must be determined with consideration for the impact of the test upon the power system. Field tests fall primarily into four categories:

a) Testing immediately following the installation of new equipment or following the repair of faulty equipment

b) Testing to determine whether equipment meets the user's specifications

c) Testing to determine parameters for use in models or to validate model study results

d) Testing to diagnose existing or impending problems

Acceptance tests are performed to demonstrate that the manufacturer has met the user's specifications. They may be performed at the installation site or in the factory. Acceptance tests are usually performed on only one unit when several units are installed under one contract. They are closely associated with modeling tests. Some tests are impractical to perform under certain conditions, and model study results or calculations should be used to obtain the required information.

Modeling tests are performed to determine parameters for model studies and to validate results from model studies. Whenever large signal testing is performed on machines connected to the power system, it is beneficial to record as much information as possible about the operating conditions of the surrounding power system. This "snapshot" of power system quantities can provide the basis for the load flow program solution upon which the transient analysis is based.

When specified, some large signal performance tests are required to be performed under rated load conditions. When possible, it is preferred that tests be performed with the synchronous machine at partial load or open-circuited, thereby minimizing or eliminating any impact upon the power system. A combination of simulations or calculations may then be used to establish the performance under the specified conditions.

Testing associated with large signal modeling has different requirements depending on the type of excitation system being tested. Since most models are used for power system dynamic simulation studies, it is important that the models represent the excitation system performance under faulted conditions. These

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conditions typically cannot be directly tested on an in-service machine. It is particularly important to ensure that tests and calculations are designed to validate the performance under fault conditions, and at ceilings and limits. Testing for acceptance and modeling purposes should be closely coordinated. Model results can be used where field testing is not practical to predict the actual operation.

Field tests to evaluate the excitation system performance and to check the operation of the excitation system functions, including limiter and protection devices, should be conducted periodically. Coordination of excitation limiter and protection functions with protective relaying settings should also be verified.

8.3.1 Large signal dynamic responses

Ceiling voltage, voltage response time, and nominal response may all be determined from an excitation system large signal dynamic response. To determine these values, it is beneficial to obtain a record of excitation control system variables for a short duration change in synchronous machine regulator error voltage corresponding to a specified drop in terminal voltage.

Transducers are available to convert terminal voltage and synchronous machine field voltage to low-voltage isolated dc signals that are compatible with recorders. Event recorders or data acquisition systems with four or more channels can be used to record transducer signals representing the synchronous machine regulator error voltage and the synchronous machine field current, field voltage, and terminal voltage. Digital excitation systems often include internal data recorders and test signal generators to facilitate these tests.

A typical dynamic response is obtained with the exciter voltage initially at rated field voltage of the synchronous machine to which the excitation system is applied. Some means is used to cause the synchronous machine regulator error voltage to change for a short duration corresponding to a specified decrease in terminal voltage. The duration of the change must be short enough that excitation control system and generator quantities remain within acceptable ranges. For high initial response excitation systems, the error signal needs to be applied for only a very short time (0.05 s to 0.2 s) due to the speed of these systems. All excitation control system stabilizing circuits (other than the power system stabilizer circuits) or other feedback control circuits that are part of the synchronous machine regulator should be adjusted as in actual operation, where possible and practical. The power system stabilizer should not be in operation for this test unless it is required for system stability purposes. Figure 13 shows the results of a field test made on one high initial response excitation system to determine ceiling voltage and excitation system voltage response time. It is representative of the type of records used but is not representative of all high initial response systems.

These dynamic response records can usually be obtained near rated operation without endangering the operation of the synchronous machine or the power system. It is essential that the duration of the change in synchronous machine regulator error voltage be limited to a short time. The time should be short enough that the change in excitation system and synchronous machine variables is acceptable, but long enough that ceiling voltage is reached.

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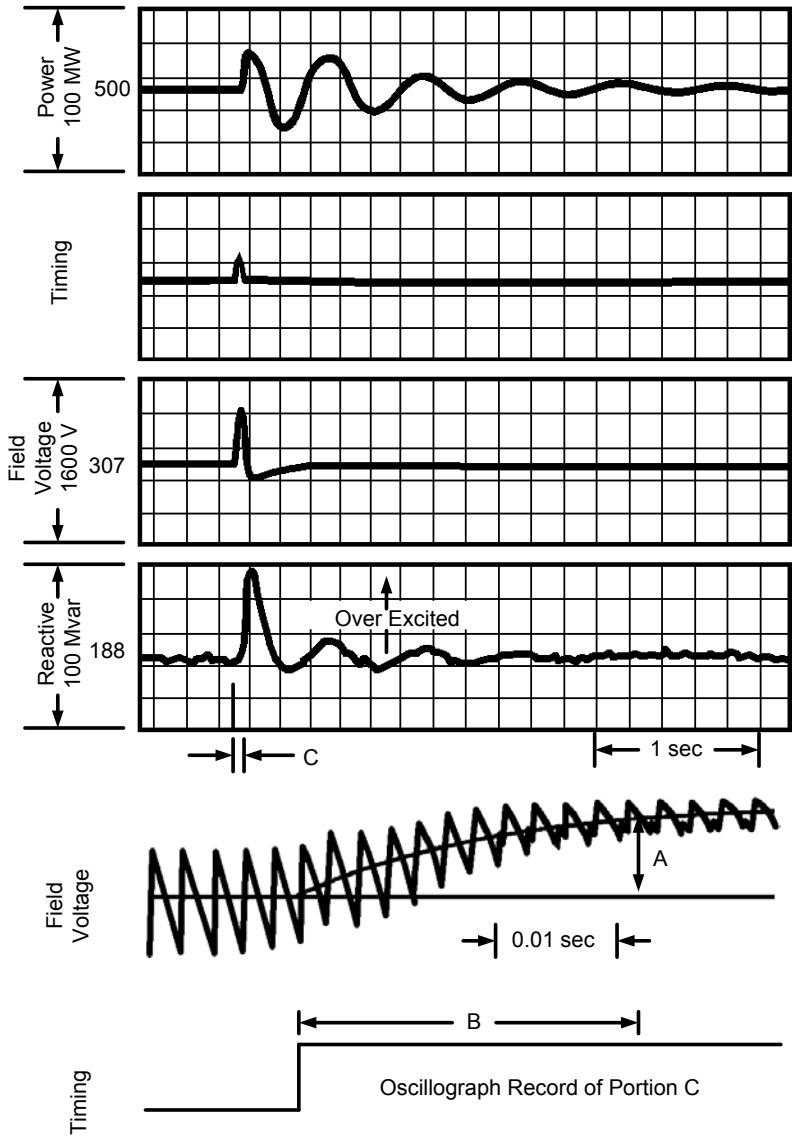

LEGEND
A = Ninety-five percent of the difference between ceiling voltage and rated field voltage
B = Excitation system voltage response time

Figure 13—Large signal (10% voltage error) field test of a high initial response excitation system

Dynamic responses are frequently obtained by one of two methods; either a test signal is injected at the summing junction where the error signal is produced, or the terminal voltage feedback signal into the sensing circuit is forced lower to reflect the desired change in terminal voltage. In either case the input should approximate a step change so an accurate response is obtained. The method of injecting a signal into the summing junction is sometimes easier to implement; but it has one drawback. Injection at the summing junction bypasses the terminal voltage transducer time constant which may be a major time constant in a fast excitation system. Analytical means may be required to combine the effects of the transducer filter with the dynamic response test results.

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8.3.2 Limiter and protection considerations

Large signal testing with the synchronous machine connected to the power system requires consideration of excitation limiter and protection settings. Large signal testing may also be used to check the operation and setting of these devices. Limiter functions are not normally tested at their final in-service settings initially upon the installation of new equipment, because a failure to operate could result in equipment damage. Typically, both the limiter and machine operating set points (e.g., current, voltage, power, etc.) are reduced to lower levels and excitation system and synchronous machine output are forced into each limiter's operating characteristic. Protective devices are typically tested by applying voltages and currents from a test set. After correct operation of the limiters and protection is verified they may be tested at their final in-service settings to verify coordination.

In some cases, limiters may interfere with the ability of the excitation system to reach ceiling current. If so, the important information for power system analysis and modeling is the limiter characteristic rather than specified ceiling current. Any runback or inverse-time type limiter characteristic that reduces field current below ceiling is also important for system modeling and should be documented during excitation system tests, if possible. (There may be a time delay from a few seconds to tens of seconds associated with this limiting function.)

8.3.3 Ceiling voltage

The measurement of ceiling voltage with the synchronous machine open-circuited will prove straightforward, except in the case of an excitation system that utilizes both synchronous machine voltage and current for the excitation system source. Brushless excitation systems that do not use slip rings may be more difficult to test unless a test equipment system employing special voltage test probes is implemented.

The measurement of a ceiling voltage at any synchronous machine load is only possible when the excitation system voltage can reach the ceiling value and return to the initial value in a duration sufficiently short that the other excitation control system and synchronous machine quantities remain within acceptable limits for plant and power system operations. When ceiling voltage cannot be reached while operating at rated field current due to operational constraints, data may be extracted from dynamic responses that produce conditions as near to ceiling as possible. These data and other information from open-circuit and reduced-load tests can generally be used to verify model parameters, and the model may be used to determine the approximate value of the ceiling voltage.

The definition of ceiling voltage is used to evaluate the forcing capability of the excitation system toward ceiling current. Since fast-acting current limiter circuits may be used in some exciter designs, the ceiling voltage must be determined before such limiter action occurs.

8.3.4 Excitation system voltage response time

Excitation system voltage response time can be determined from the excitation system voltage dynamic response (see 8.3.1). The time measurement begins at the time the disturbance is initiated and ends when the dynamic response reaches 95% of the difference between ceiling voltage and rated field voltage.

Where the synchronous machine field voltage record of the dynamic response does not attain ceiling voltage, analytical means may be used to determine the approximate voltage response time. The excitation system voltage dynamic response should come as close to ceiling voltage as possible or practical for the analytic means to be most accurate.

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8.3.5 Nominal response

The nominal response is based upon a linear approximation of the excitation system voltage large signal dynamic response for the first 0.5 s. A straight line is determined such that the area under the line and the area under the dynamic response are the same over the time interval of interest. In Figure 2, the line ac is chosen to make these areas equal. The excitation system voltage at 0.5 s on this line (point c in Figure 2) is used to calculate the nominal response. The increase in excitation system voltage (ce − ao) is divided by the rated field voltage (ao) and by the time interval (oe = 0.5 s) to obtain the nominal response in per unit rated field voltage per second. (Note that this is the only excitation system criterion that is measured in per unit rated field voltage. All other excitation system constants are referenced to one per unit air gap voltage. This is particularly important for modeling.)

The determination of nominal response requires that the excitation system voltage be initially equal to the rated field voltage of the synchronous machine. During field tests, it may not be possible to operate with this initial field voltage due to power system voltage and/or reactive power restrictions. For large exciters, factory tests may prove impractical due to the large power magnitudes that are involved. If the specified conditions cannot be met, data from open-circuit and reduced-load tests can be used to approximate the nominal response.

8.3.6 Ceiling current

Field testing of ceiling current may be impractical because of the excessively high current that is required. The impact on the power system from this disturbance may be unacceptable. Operation of another nearby unit to counteract the effects of the unit under test may alleviate these concerns. When the field current is limited to some predetermined value by the use of an instantaneous current limiter, tests may be performed to determine that the limiter is functioning properly. It may be necessary to test the limiter operation at a current value considerably below the desired value in order to limit the impact on the power system. Analytical means may then be used to determine the desired limiter settings from the test results at the lower current value.

8.4 Fault testing

Fault tests where three-phase, line-to-line, or line-to-ground faults are used for testing the operation of transmission line or synchronous machine protection also provide a good opportunity for evaluating the performance of the excitation control system under fault conditions. Records from these tests provide a method for obtaining values for the large signal performance criteria under conditions more closely approximating typical fault conditions, including reduced terminal voltage, increased stator current, and induced rotor currents. These responses may not match those obtained under other conditions; however, modeling programs should be able to produce matching results.

8.5 Power system disturbances

Records obtained during power system disturbances are valuable for verification of model parameters. When disturbance monitoring equipment is being installed, consideration should be given to excitation control system quantities of interest. Terminal voltage and field voltage of the synchronous machine should be the highest priority for monitoring.

Consideration should be given to monitoring transduced signals (i.e., the controller's internal intermediate signals such as voltage error and AVR output) rather than the ac waveforms. If the excitation system equipment provides transduced signals for field current, synchronous machine regulator output and/or the regulator error signal, it may be beneficial to monitor these signals as well. Where equipment is available,

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monitoring of the excitation system stabilizer, the power system stabilizer, and over and under excitation limiter operation should be considered. For maximum benefit, the recording equipment bandwidth must be appropriate for the signals being monitored. Modern digital excitation systems often contain internal data recorders that are capable of triggered disturbance monitoring of excitation and generator quantities. Where available, these should be configured to capture power system disturbances as well as excitation system problems.

9. Small signal performance testing

9.1 General

Small signal performance tests of the overall excitation control system are used to demonstrate that the manufacturer has met the specified requirements of the customer (acceptance testing), to tune the excitation system for the desired response to meet power system needs (initial and periodic adjustment testing), to determine parameters for models used to predict small signal performance (model development testing), and to ensure that the equipment is working correctly (periodic maintenance testing). Field tests to evaluate the excitation system performance and to check the operation of the excitation system functions, including limiter and protection devices, should be conducted regularly based on plant maintenance schedules or as required by local power system regulatory entities. Changes to power system topology in close proximity to the power plant may justify more frequent testing. Additionally, coordination of excitation system limiter and protection functions with protective relaying settings may be verified during these tests.

9.2 Excitation control system components

In addition to the performance of the overall excitation control system, transfer functions describing the dynamic performance of individual components or a group of excitation control system components can be derived from small signal performance data of various elements. Data may be obtained by time domain dynamic response testing or by frequency response testing. One of the important uses of the transfer functions obtained from these tests is in the development of models for computer simulation of power system dynamics (see IEEE Std 421.5).

The major components of an excitation system to be characterized include the power amplifier (rotating or static), the voltage regulator terminal voltage transducer and summing junction, and the voltage regulator stabilizing means. Additionally, modeling data may be desired for the PSS and excitation limiters.

9.2.1 Small signal dynamic response tests

Dynamic response testing consists of applying a signal (typically a step) into the input of the element or elements under test and recording the output response. For example, rise time, overshoot, and settling time can be obtained directly from the dynamic response test. Another direct use of the dynamic response test is in periodic verification of performance. This can be done by comparing the result of the transient test performed periodically with the corresponding tests performed at the time of unit commissioning. This procedure is sometimes called *signature analysis*. An example of dynamic response tests in signature analysis is given in 9.4.

A disadvantage of dynamic response testing is that if the transfer function of the element under test is to be determined, the data must be reduced to a form that will yield this information. Methods of accomplishing this include performing an inverse Fourier transform on the dynamic response and the test signal, usually with the aid of a digital computer; performing a graphical analysis on the data approximating the dynamic

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response as a series of impulse responses; and iterative simulation on a digital computer. All of these methods can be lengthy and involve extensive iterative calculations. Also, accuracy may be questionable if the output of the element being tested contains appreciable noise.

If the system being tested contains a dominant real pole or a dominant pair of complex poles (near the $j\omega$ axis in the s plane), a dynamic response test is often useful in measuring the position of these s plane poles. For example, if a damped sinusoidal oscillation dominates the response, a measurement of the oscillation frequency of the response (f_o in Hz) and the time constant (τ_o in seconds) that its envelope decays at is enough information to locate these complex poles on the s plane. The position $\sigma \pm j\omega$ will be at $\sigma = -1/\tau_o$ (sec^{-1}) and $\omega = 2\pi f_o$ (radians per second). Digital excitation systems often contain built in data recorders and step signal generators to facilitate such testing and tuning.

9.2.2 Frequency response testing

Frequency response testing consists of applying a known driving signal to the system (or a portion of the system) and measuring the output (or an intermediate point) with respect to the input. There are several methods which are used to measure frequency response, including methods that apply either sinusoidal or noise signals. Frequency response testing has the advantage that the transfer function of the element under test is often immediately evident. For this reason, the frequency response method is recommended for determining the small signal transfer functions of excitation system components. Frequency response analyzers can directly produce the gain and phase plots shown in Figure 5 (Farmer and Agrawal [B18]). Some analyzers use averaging techniques to reduce the effects of noise and thus may require only a relatively small driving signal. This may be useful for on-line tests to minimize the impact of testing on the power system.

Digital excitation systems will require analog-to-digital inputs and digital-to-analog outputs with sufficient bandwidth and resolution to permit frequency response testing. Input-to-output characteristics of these interfaces should be tested at the same time the response measurements of the excitation system are made.

Digital excitation systems often include integrated data recording facilities to aid the testing and documentation process.

The following precautions should be observed in frequency response testing.

a) Care should be taken to ensure that the driving signal does not cause saturation of the elements under test. A signal magnitude causing output variations that are linear about the quiescent operating point is necessary.

b) The range of test frequencies should be broad enough to fully describe the dynamic characteristics of the element(s) under test. The frequency resolution should accurately describe abrupt changes in the response curve, particularly in the vicinity of resonant peaks. With most analyzers, it is possible to cover the complete range of interest in a single measurement. (Example frequency range: 0.05 Hz to 20 Hz, but local regulatory requirements may apply.)

c) In measuring the response of rotating exciters in the factory, it may be necessary to insert a power amplifier between the driving signal source and the exciter field. The transfer function of the power amplifier should be recorded.

d) In measuring the frequency response of the excitation system elements it may be necessary to insert isolation transducers between the output of the power component and the test instrumentation. Also, it may be necessary to filter the output of the power component. The transfer functions of all transducers and filters used should be recorded.

e) The test should be performed in the expected operating range of the element under test. For example, the frequency response of a rotating amplifier varies with loading. When measuring the response of individual elements of an excitation system, care should be taken with respect to

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possible variation of expected overall response when the elements are connected together due to loading and interaction effects. This applies primarily to excitation systems employing passive circuit elements and rotating or magnetic amplifiers. In excitation systems employing solid-state control elements, those elements may be free of interaction and loading effects. Should these effects be present, the overall frequency response of the group of elements interconnected in their normal configuration should be measured in addition to the response of the individual elements. Typical frequency response characteristics and corresponding transfer functions of elements commonly encountered in excitation systems are shown in Annex B.

9.3 Field testing

Time domain and frequency response characteristics can fully describe the small signal dynamic performance of an excitation control system. These characteristics can be determined by field testing as described in this section.

With the exception of those alternator rectifier excitation systems in which the exciter terminals are not available, and compound rectifier excitation systems, most of the excitation system component tests described in 9.2 could be performed at the factory or in the field prior to testing of the overall excitation control system.

Most excitation control systems include some form of internal stabilization, such as rate feedback, transient gain reduction/increase, proportional-integral (PI), or proportional-integral-derivative (PID) control. The function of this internal compensation is to ensure stability of the excitation control system with the synchronous machine off-line, and to provide a means of adjusting the dynamic response of the system. Many newer excitation systems allow for the use of multiple structures and settings to meet the changing requirements after the unit is synchronized to the power system and when it is operated at higher loads.

During startup of a new synchronous machine installation, it is normal procedure to check the dynamic response of the unit. This is best done by inserting a small step change into either the reference or sensing circuitry of the voltage regulator and recording the machine terminal voltage response. It is recommended that a wide bandwidth ac to dc voltage transducer with adjustable null balance providing zero suppression be used to record only the deviation in terminal voltage from its steady-state level.

For an excitation system having small time constants, an acceptable dynamic response might be considered as one having no more than two overshoots with maximum overshoot of 5% to 15%. In some applications, it may be desirable to increase the bandwidth of the excitation system for faster response. In such applications, higher overshoot can occur.

Field testing to obtain the frequency response characteristics of excitation control systems yields data which is useful for system identification and for verification of mathematical models for use in power system stability studies. Data obtained from frequency response testing is also useful in determining suitable settings for power system stabilizers as described in 9.4. In measuring the frequency response of an excitation control system, a small signal, typically limited to cause less than +/−2% excursions in terminal voltage at low frequency, is inserted into the regulator reference and sensing summing junction. The excitation control system closed-loop frequency response should be measured with a minimum bandwidth from 0.1 Hz to 3 Hz. It is frequently desirable to record deviations in the variables of interest during these tests.

Phase and amplitude responses can be obtained directly using a multi-channel analyzer. To obtain a 0.25% to 2% variation in terminal voltage will require a progressively larger driving signal at higher frequencies. Likewise, in measuring frequency response using noise-based techniques, the power level and bandwidth of the noise driving signal must be carefully selected to provide adequate power for measurements without permitting excessive transients. Care should be taken to ensure that the excitation control system quantities do not encounter limits. Thus, they should be monitored during the test. Preferably, frequency response

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measurements should be made both with the unit off-line and with the unit on-line operating near full load with external power system conditions as in normal operation. If a certain transmission configuration is suspected to stimulate oscillations, additional tests may be performed with this configuration, if possible.

CAUTION

Some common frequency response measurement devices operate by injecting a fixed frequency exciting signal, either automatically swept or discretely switched during the measurement. This approach has the advantage that it requires little or no averaging to arrive at a reasonable measurement, and it allows the user to directly control the energy injected at each frequency. If this method is employed at least the following precautions should be observed:

a) When performing a frequency response test on a generator connected to the grid, caution should be exercised when injecting frequencies that are close to the resonant frequencies of the machine (e.g., local mode, interarea mode, intraplant mode) or neighboring machines.

b) Extreme care should be taken when injecting frequencies higher than 3 Hz, as these may correspond to the lowest shaft torsional frequencies of turbo generator sets. The turbine manufacturer should be consulted to obtain a torsional profile of the rotor-turbine shaft prior to proceeding with testing.

CAUTION

Failure to employ these and other appropriate precautions may result in damage to the turbine or generator shaft which could require costly repair or replacement and cause an outage of the power generation equipment.

In general, the data obtained when operating the synchronous machine near full load and exciting the unit near mechanical or electro-mechanical resonant frequencies is not reflective of the closed-loop voltage regulator loop frequency response since it is affected by the rotor angle variations. Therefore the range of frequencies around the critical modes may be bypassed, and measured points at other frequencies may be compared against theoretical values to develop a full spectrum model. Some frequency response analyzer hardware provides provisions for bypassing a range of frequencies or provides automatic signal source output control based on one of the measured quantities. Alternately, data for these frequency ranges can be obtained by operating the synchronous machine at a lower power output level where excitation changes do not result in significant torque changes (Larsen and Swann [B32]).

As previously described in 6.3.1, open-loop frequency response characteristics are useful in determining relative stability of a feedback control system; however, it is usually not practical to measure the open-loop frequency response characteristics of excitation control systems directly by field testing. They can, however, be derived by superposition of the synchronous machine regulator characteristics with those of the combined exciter and possible synchronous machine, if the regulator error signal can be measured.

If the regulator error signal is not available, it may be possible to derive the open-loop frequency response from measurements obtained from the closed-loop feedback control system. If the closed-loop frequency response $\Delta V_T/\Delta V_R$ (see Figure 14) is easily measurable by test, but the open-loop response $G(s) = \Delta V_T/\Delta e$ is not, G(s) can be derived from the results of the closed-loop frequency response as follows:

Let $G(s) = \Delta V_T/\Delta e$ (2)

Since $\Delta e = (\Delta V_R - \Delta V_T)$, (3)

then

$$\Delta V_T/(\Delta V_R - \Delta V_T) = (\Delta V_T/\Delta V_R)/[1 - (\Delta V_T/\Delta V_R)] \quad (4)$$

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Note that the closed-loop frequency response $\Delta V_T/\Delta V_R$ must be expressed as a phasor quantity, having both a magnitude and phase angle. Inaccuracies may occur in this method, particularly at very low frequencies where Δe may be very small.

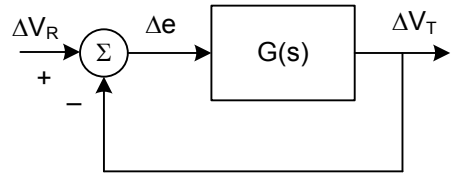

Figure 14—Block diagram of a closed-loop feedback control system

9.4 Field testing with power system stabilizers

Testing methods for tuning a PSS while connected to the power system may be subject to local regulations.

Frequency response testing is a convenient means for experimentally checking parameter ranges, determining parameter settings, and verifying PSS performance. The PSS control signal transducer calibration must be determined. Analog transducers should be calibrated in volts/pu where 1 per unit is based upon rated speed, frequency, or MVA depending on the stabilizer input signal. Depending on design and configuration, transducer response time may, in some cases, be determined experimentally by frequency response techniques similar to those described earlier for testing of the components of the excitation system.

The settings of the time constants of the phase compensating lead-lag functions of the stabilizer can be determined by several techniques once the frequency response of the excitation control system has been measured (as described in 7.2). Regardless of the technique used to determine these settings, the ultimate goal is to compensate for the phase lags in the excitation control system so that the stabilizer, through the excitation system, adds a component of torque which is in phase with speed deviation (DeMello and Concordia [B16]).

The stabilizer should ideally increase the damping component of torque without affecting the synchronizing component. For the lower frequencies of oscillation associated with interarea modes, adjusting the terminal voltage variation to be in phase with the frequency deviation will fulfill the same requirement. Desired alignment of the phase compensating network may be verified by frequency response measurements. Typical frequency responses of lead lag functions are presented in Figure C.4 and Figure C.5.

The PSS gain should be maximized within the constraints imposed by the recommended stability margin of the control loop. The maximum permissible gain depends upon many factors and can be determined by test. Selection of the in-service PSS gain for a given installation usually involves balancing conflicting requirements. As PSS gain is increased, the damping contribution to electro-mechanical modes (both interarea and local) will increase.

On the other hand, increasing gain will tend to reduce the damping of control-loop “exciter” modes. Some local power system coordinating entities may establish the appropriate gain by attempting to maximize the damping of all modes or by setting a minimum damping requirement for all critical modes. In other areas, the damping of observable modes (e.g., local and exciter) is used during test to establish an appropriate gain. Other factors such as large signal performance and the relationship with PSS and exciter limits should be considered in establishing the final gain.

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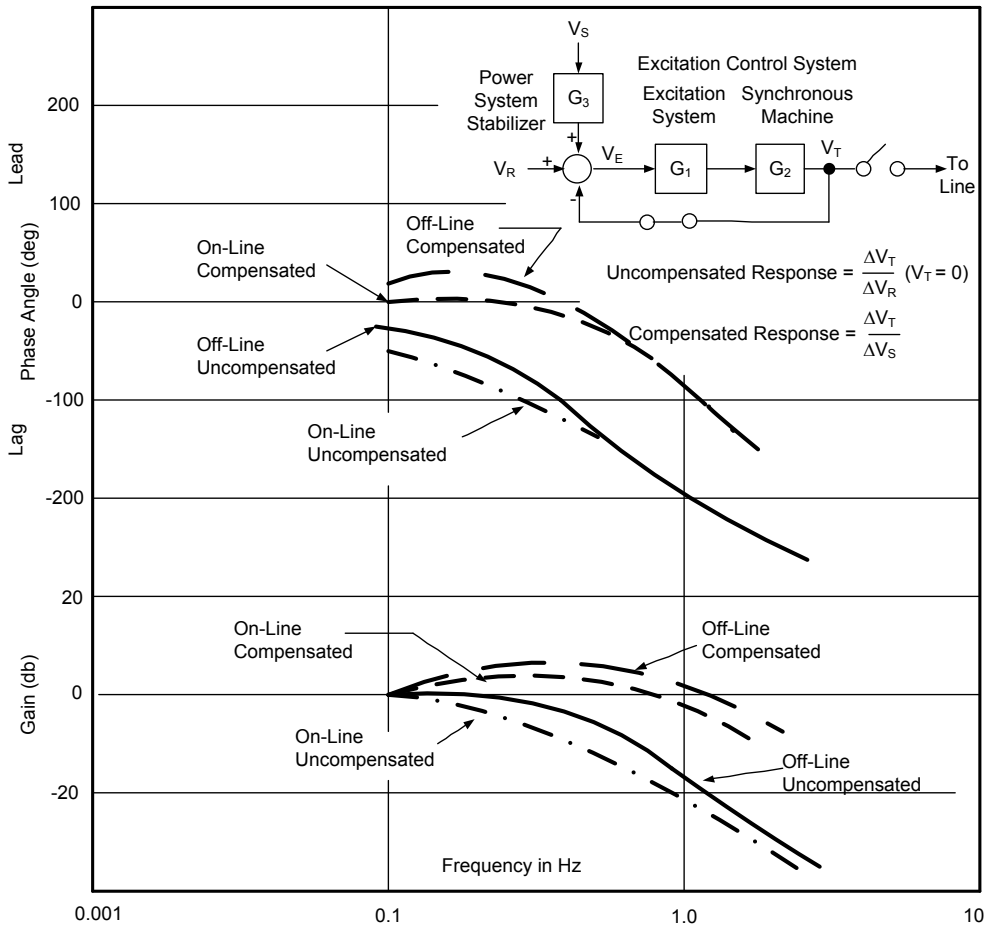

Figure 15—Frequency response of a typical excitation control system (with dc commutator exciter) with and without a power system stabilizer

Figure 15 shows typical frequency response data of an excitation control system with and without a power system stabilizer. Note that the frequency response characteristics ΔVT/ΔVR and ΔVT/ΔVS shown are convenient to measure and yield adequate information for determining and verifying suitable settings of the PSS to improve damping of the lower frequency (< 0.7 Hz) interarea oscillations. In determining and verifying suitable PSS settings to improve damping of higher frequency (1 Hz to 2 Hz) local mode oscillations, the transfer function for which compensation should be provided is ΔTe/ΔVR, which is the change in electrical torque (or power) with respect to change in reference voltage. Actual measurement of this transfer function may also be useful.

If adjustments have been adequate, the phase angle of the on-line compensated response will be near zero over the useful range as shown in Figure 15. The gain curve will also be reasonably flat over this range. Readjustments of time constants may be made and the frequency response test rerun, if necessary. Note that with the gain curve nearly flat, unlike in the initial frequency response run, it will not be necessary to increase the input signal as the frequency is increased. However, care will again be necessary to ensure that signal limiting does not occur at the higher frequencies.

Verification of power system stabilizer dynamic performance can also be made by dynamic response testing. The field test data in Figure 16 show the dynamic response of an excitation control system equipped with a power system stabilizer to a step change in regulator input signal.

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A recording of dynamic response such as that shown in Figure 16 is also very useful in signature analysis; it provides a means of periodically verifying that a power system stabilizer is functioning properly. Dynamic response characteristics (the "signature") of an excitation control system with and without the power system stabilizer can be quickly obtained periodically and compared to the data recorded during the initial operation of the particular equipment. Similar results can also be obtained using frequency response measurements.

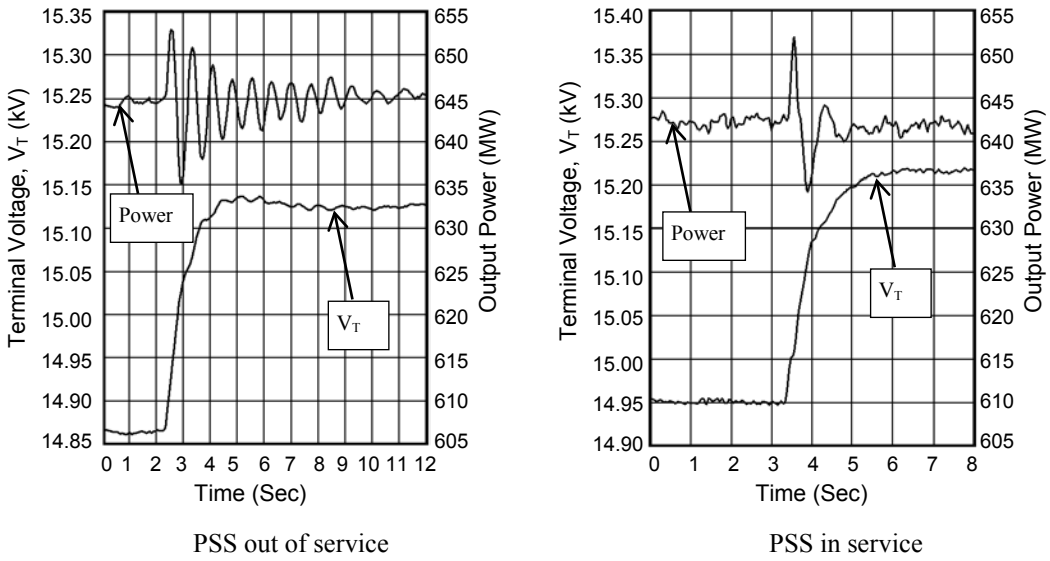

Figure 16—Field test data—power system stabilizer signature tests

9.5 Special considerations for digital systems

Digital systems may have the following features to facilitate testing:

a) Accessible memory to allow data variables to be recorded with the ability to trigger records and select quantities to be recorded, sampling rate, and record length

b) A user-assigned analog/digital interface to inject external test signals or triggers at various points within the digitally implemented function blocks

c) A user-assigned digital/analog interface to provide external access to calculated quantities or other means to measure these quantities

d) Internal test functions that may be programmed to introduce step or other test signals (e.g., sinusoid, noise, etc.)

9.6 Testing reactive load sharing and line drop compensation

Reactive compensation—sometimes named reactive current compensation (RCC), reactive differential compensation (RDC), or line drop compensation (LDC)—may be tested by performing voltage regulator reference step-response tests on open circuit and on-line. For a generator with a dedicated transformer connection to the high-voltage system and no form of AVR connection to other generators, the procedure is as follows:

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With the generator synchronized at no (or minimum) load and rated terminal voltage, tabulate terminal voltage and reactive power. With the reactive compensation setting, X_C, set to zero (or with the unit on open circuit), measure the change in both terminal voltage and reactive power (if applicable) while inserting a measured step change into the terminal voltage reference. Select the final value of X_C, reinsert the step change, and confirm the calibration from Equation (5).

$$\Delta E_T = \Delta E_{T\text{-}REF} - X_C \, \Delta Q \tag{5}$$

where

ΔE_T is the per-unit change in RMS terminal voltage = $(E_{Tfinal} - E_{Tinitial})$/rated kV

$\Delta E_{T\text{-}REF}$ is the per-unit change in terminal voltage reference (equals terminal voltage change on open circuit or when reactive compensation set to zero)

ΔQ is the per-unit change in reactive power = $(Q_{final} - Q_{initial})$/pu MVA

X_C is the reactive compensation expressed in pu ET_{REF}/pu MVA (X_C is positive for reactive droop and negative for transformer or line-drop compensation)

For two generators tied together at their terminals sharing a step-up transformer using reactive differential compensation (also referred to as cross-current compensation), first perform the tests as described above on each individual unit; with the second unit shut down, on manual regulator, or with its cross-current compensation out of service (being careful to not take the reactive compensation off both generators). Then monitor both units while performing a step change in one generator's voltage regulator reference.

The value of the reactive differential compensation may be found from:

$\Delta E_{T1} = \Delta E_{T\text{-}REF1} - X_{C1} \times \Delta Q_1 - RDC_1 \times \Delta Q_2$ (for the generator with the step input inserted)

$0 = X_{C2} \times \Delta Q_2 + RDC_2 \times \Delta Q_1$ (for the generator reacting to the adjacent generator)

where

X_{C1} is the reactive compensation setting for generator with the step inserted

X_{C2} is the reactive compensation setting for the adjacent generator

RDC_1 is the reactive differential compensation setting on the generator with the step inserted to account for the reactive load of the adjacent generator

RDC_2 is the reactive differential compensation setting on the adjacent generator to account for the reactive load of the generator with the step inserted

9.7 Testing power factor and var controllers

These controls may be tested in the same way as the AVR, using a step-change of voltage reference or a step-change of controlled variable reference (e.g., pf, var). The reference change, terminal voltage, controlled variable, and any supervisory functions (e.g., voltage) should be monitored. The control deadband should also be measured. Gradually increase the applied step size until the control operates. Performing step tests in both increasing and decreasing directions should result in different final values equal to the deadband. Since the controller is a supplemental loop, its response is expected to be a slow integrator to allow for the voltage regulator to be initially responsive to system disturbances.

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Annex A

(informative)

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Excitation system performance and modeling

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Annex B

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Basis underlying the concept of nominal response

B.1 Explanation

In order to appreciate the usefulness of the excitation system nominal response as a measure of the ability of the excitation system to affect transient stability, consider a synchronous generator with only a single rotor circuit in each axis and no saliency. The phasor diagram, which must be satisfied under both transient and steady-state conditions, is shown in Figure B.1.

Since the flux linking the rotor circuits and the stator cannot change instantly, the electrical torque can be conveniently expressed as:

$$T_e = \frac{E' E_S}{X' + X_E} \sin \beta \tag{B.1}$$

Under the assumption of "constant" flux linkages (sometimes referred to as "classical"), this torque is not a function of time, and a greatly simplified analysis is possible.

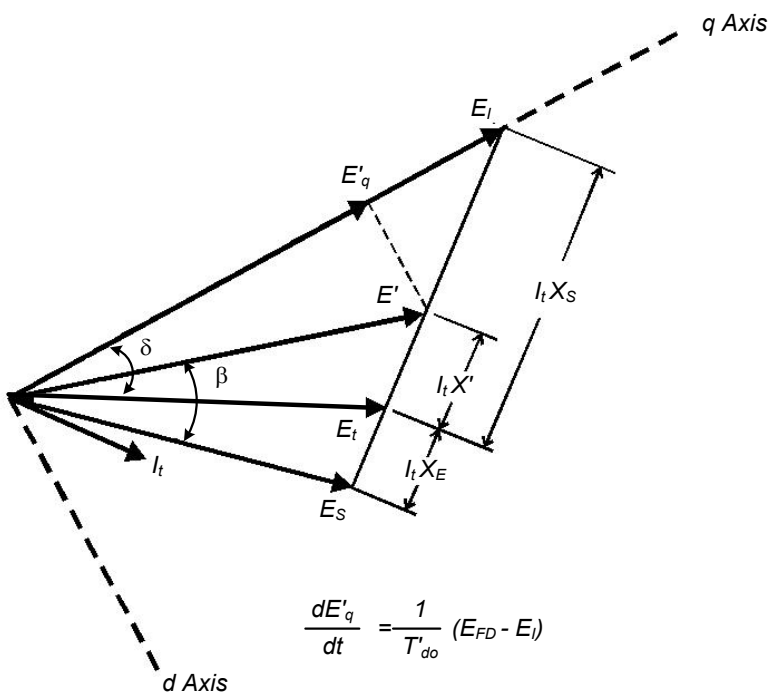

Figure B.1—Phasor diagram of a simplified synchronous generator against an infinite bus

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However, currents can change instantly, and components are induced in the rotor circuits to balance those in the stator circuit in order to satisfy the constant flux linkage theorem (see Excitation System Dynamic Characteristics [B6]). The flux linkages do change slowly and in the direct axis are described by the relationship:

$$\frac{d}{dt}E'_q = \frac{E_{FD} - E_I}{T'_{d0}} \tag{B.2}$$

$$E'_q = \frac{1}{T'_{d0}}\left[\int E_{FD}dt - \int E_I dt\right] \tag{B.3}$$

The term including E_I is determined from generator and power system characteristics. The effect of the excitation system is determined by the E_{FD} term.

Following a severe power system disturbance, the generator rotor angle undergoes large excursions at a frequency determined by the rotational inertia and the electrical stiffness of the power system. The maximum angular swing will normally peak between 0.4 s and 0.75 s after the disturbance initiation, and the excitation system must act within this time period to affect transient stability considerations. Accordingly, 0.5 s was chosen for the definition time period of the nominal response. The nominal response is the slope of a straight line (in per unit per second) constructed to encompass the same area as the actual exciter voltage curve over the 0.5 s period. It may be expressed as:

$$\int_0^{0.5} E_{FD}dt = \int_0^{0.5} (E_0 + Kt)dt \tag{B.4}$$

The area under the actual voltage-time response curve equals the area under the constructed straight line. This reduces to:

$$\text{Nominal response} = \frac{K}{E_0} = \frac{8}{E_0}\int_0^{0.5} E_{FD}dt - 4 \tag{B.5}$$

B.2 Nomenclature for Annex B

(See also Excitation System Dynamic Characteristics [B6].)

X′ = Transient reactance of the machine.

X_s = Synchronous reactance of the machine.

NOTE—The assumption is made that both the transient and synchronous reactances are constant regardless of the angular position of the rotor with respect to the armature reaction, i.e.:

$X' = X_D' = X_Q'$ and $X_S = X_D = X_Q$

This assumption is made to facilitate analysis of relatively simple cases of transient stability.

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E' = Equivalent voltage back of transient reactance, corresponding to the more slowly decaying component of flux linkages of the rotor.

E'_q = Component of E' due to the more slowly decaying component of flux linkages in the direct axis. These flux linkages generate a voltage in the quadrature axis lagging the direct axis by 90°.

E'_d = Component of E' due to the more slowly decaying component of flux linkages in the quadrature axis.

X_e = External reactance between the machine terminals and the infinite bus.

E_S = Voltage of the infinite bus.

E_I = Voltage corresponding to the generator field current (saturation neglected).

E_0 = Initial value of generator field voltage; assumed to be generator rated load field voltage unless otherwise specified.

I_t = Armature current.

T_e = Electrical torque.

T'_{d0} = Direct axis transient open-circuit time constant.

K = Slope (volts per second) of the straight line constructed to determine nominal response.

δ = Electrical angular displacement between the rotors or the voltages in quadrature with the rotors.

β = Electrical angular displacement between the infinite bus voltage and the voltage behind transient reactance.

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Annex C

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Frequency response characteristics of typical excitation control system elements

This annex consists of a series of figures that illustrate the frequency responses for several typical excitation system elements.

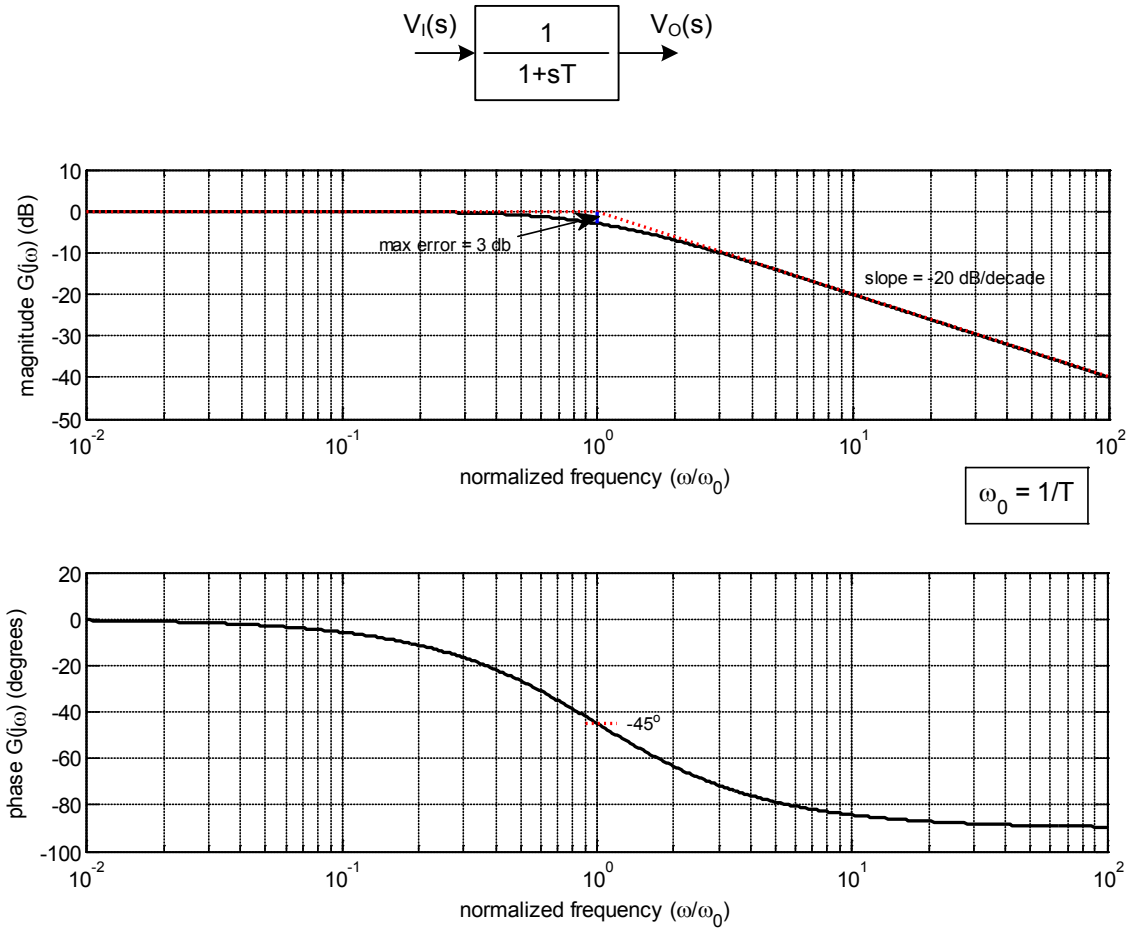

Figure C.1—First order system

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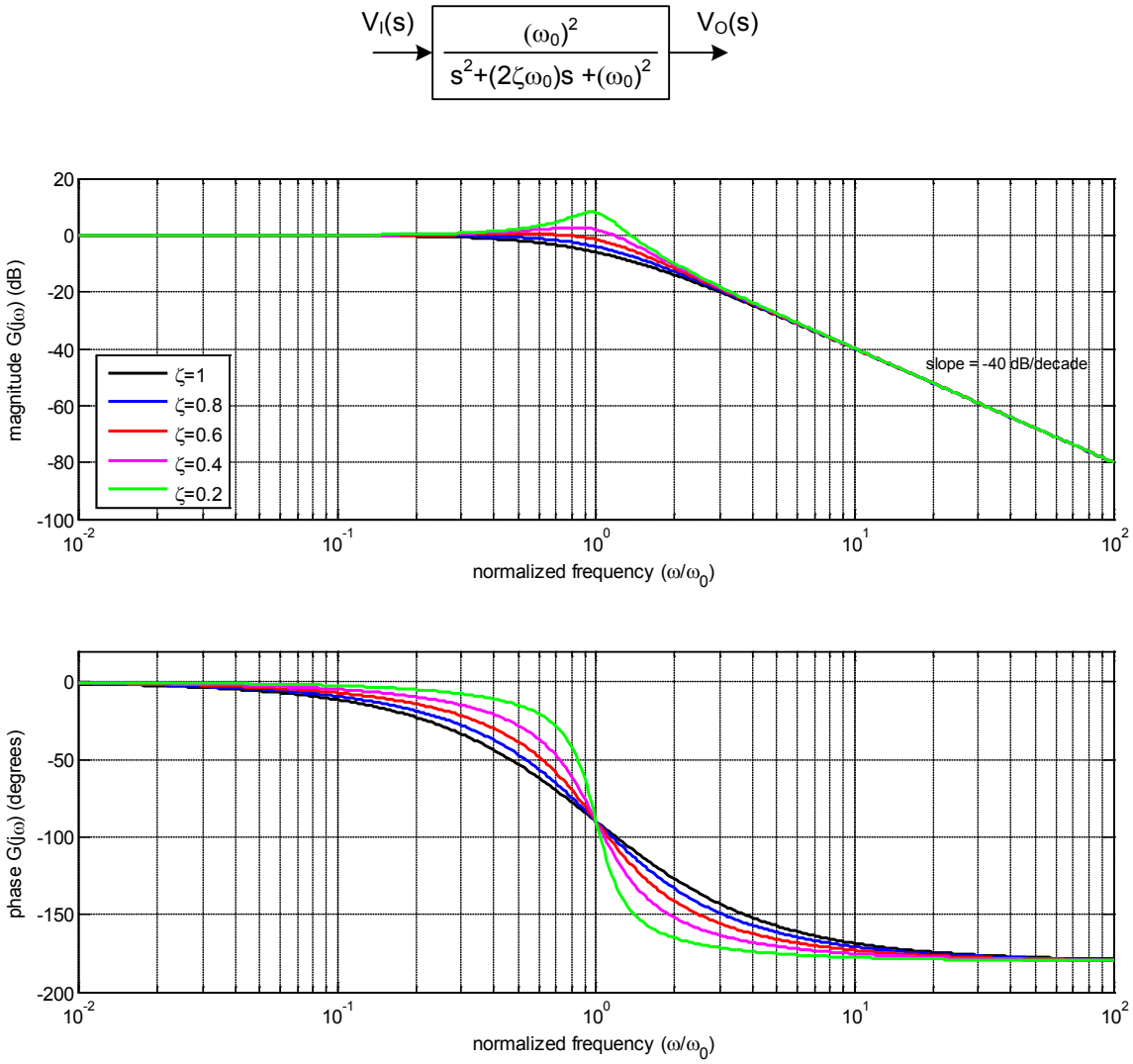

Figure C.2—Second order system

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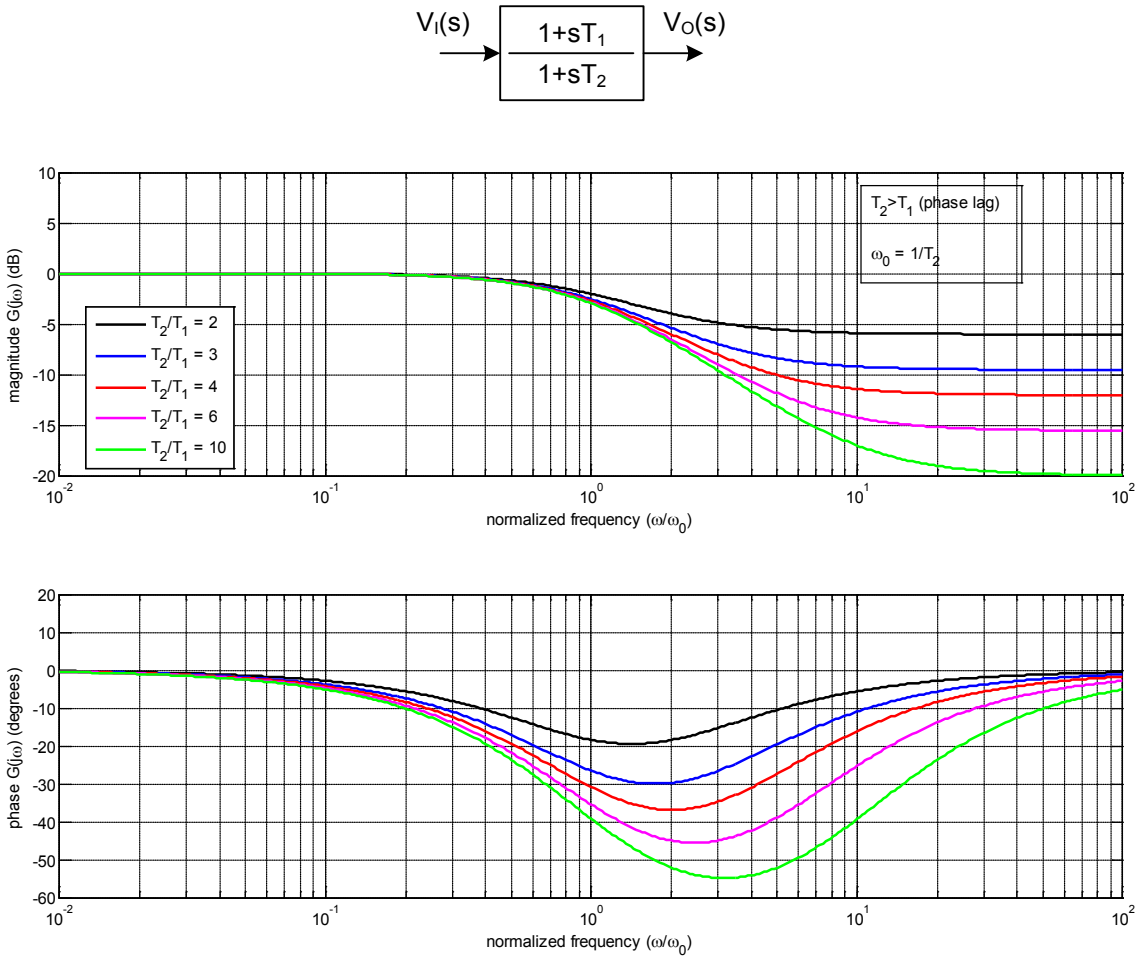

Figure C.3—Lag–lead function

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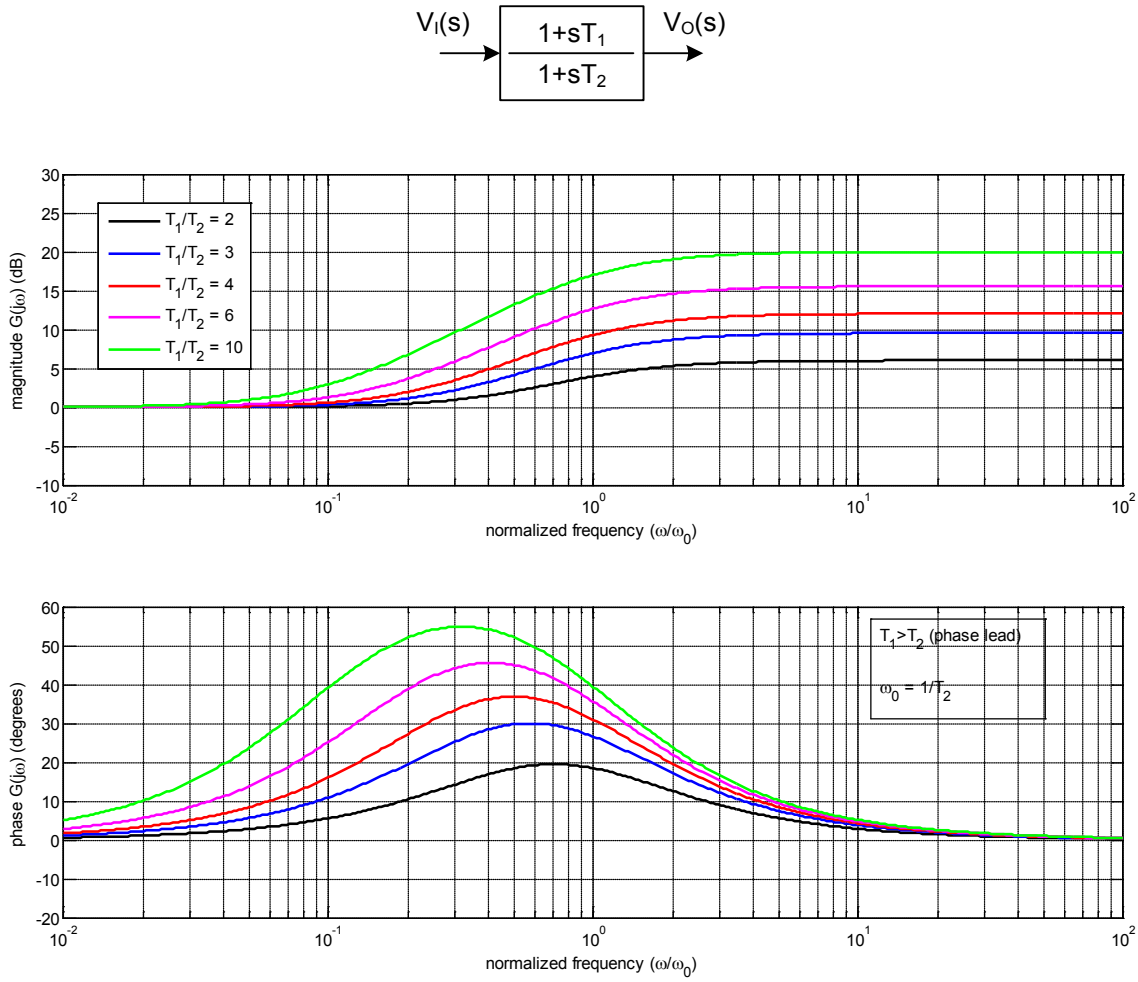

Figure C.4—Lead-lag function

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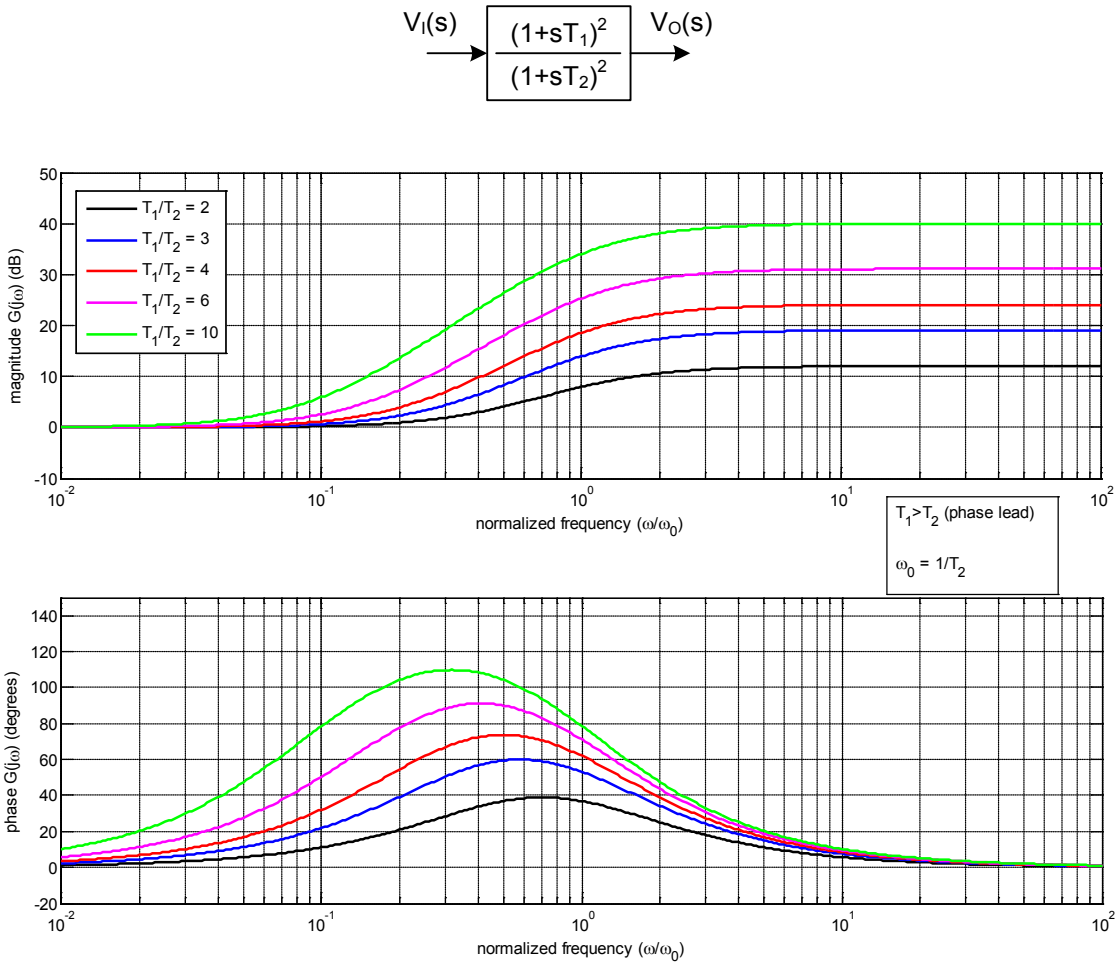

Figure C.5—Twin lead-lag function

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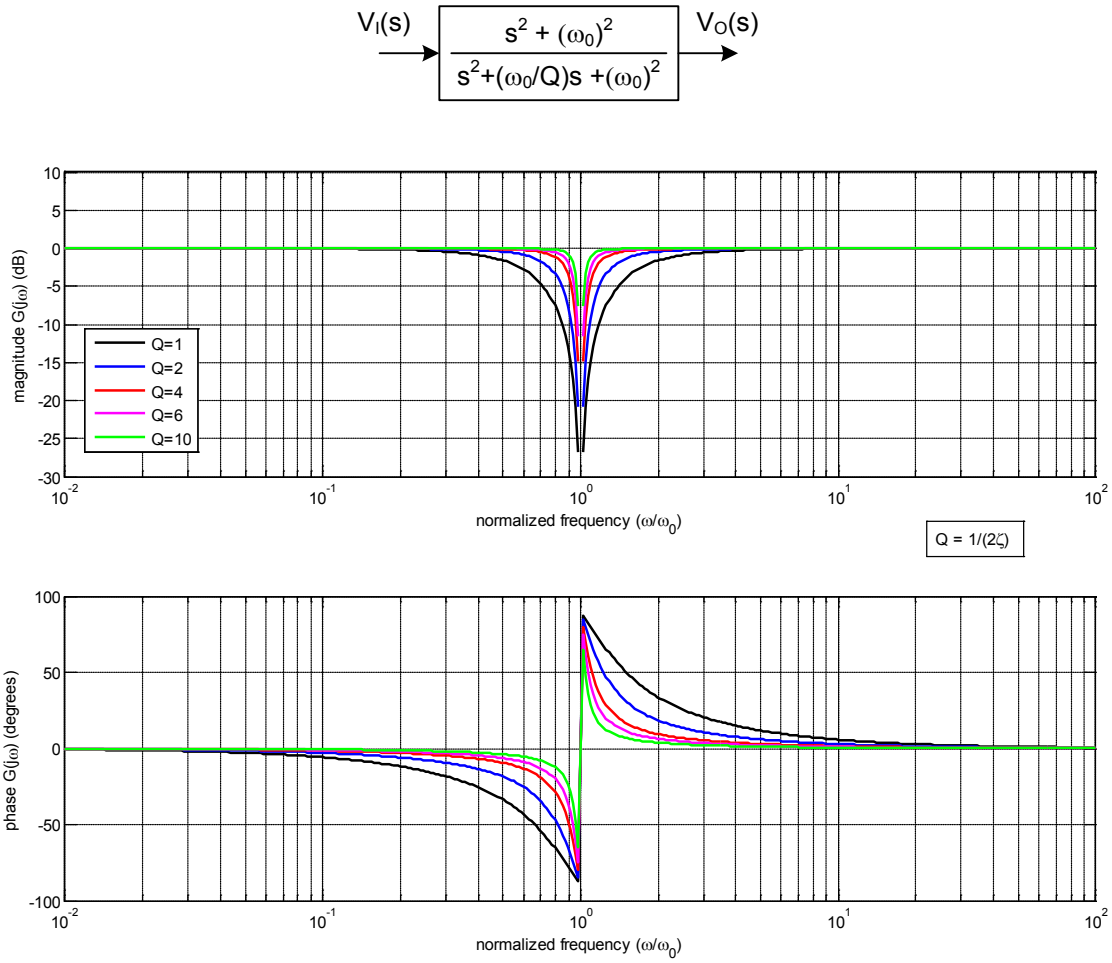

Figure C.6—Biquadratic notch filter function

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